ATTENDTRACK: A WEB-BASED EVENT ATTENDANCE AND EVALUATION SYSTEM IN LAPU-LAPU CITY COLLEGE

A Project Study

Presented to

the College of Technology

LAPU-LAPU CITY COLLEGE

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ABSTRACT

Managing event attendance through manual methods is often time-consuming, inefficient, and prone to human error, particularly in academic institutions that conduct multiple events. To address these challenges, this study developed AttendTrack, a web-based event attendance and evaluation system that utilizes QR code-based attendance scanning, real-time monitoring, automated report generation, and role-based access control. The system also includes event registration, participant management, notifications, dashboards, and evaluation features to improve the overall event management process. Developed using the Agile Software Development Framework, the system was evaluated through a descriptive quantitative research design based on the ISO/IEC 25010:2011 software quality standards with 50 respondents from Lapu-Lapu City College. The results showed very satisfactory ratings in functional suitability, performance efficiency, usability, reliability, and security. The findings indicate that AttendTrack effectively enhances attendance monitoring, improves data accuracy, reduces manual workload, and provides a reliable solution for modernizing event attendance and evaluation processes.

Keywords: AttendTrack, event attendance monitoring, QR code, web-based system, Agile Framework, ISO/IEC 25010:2011

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THE RESEARCHERS

DEDICATION

This study is wholeheartedly dedicated to the administration, faculty, and student body of Lapu-Lapu City College who continue to navigate the challenges of managing event attendance and evaluation within the institution. You are the inspiration behind the creation of ATTENDTRACK: A Web-Based Event Attendance and Evaluation System in Lapu-Lapu City College; your needs and insights guided the vision of this project.

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THE RESEARCHERS

TABLE OF CONTENTS

PAGE

APPROVAL SHEET **I**

ABSTRACT **II**

DEDICATION **IV**

TABLE OF CONTENTS **V**

LIST OF FIGURES **X**

CHAPTER 1 **1**

THE PROBLEM AND ITS SCOPE **1**

RATIONALE OF THE STUDY **1**

STATEMENT OF THE PROBLEM 3

OBJECTIVE OF THE STUDY 5

SIGNIFICANCE OF THE STUDY 6

SCOPE AND LIMITATIONS OF THE STUDY 7

SCOPE OF THE STUDY 7

LIMITATIONS OF THE STUDY 8

THEORETICAL BACKGROUND **10**

SOFTWARE DEVELOPMENT PROJECT FRAMEWORK 10

ITERATIVE DEVELOPMENT FRAMEWORK 12

REQUIREMENTS PHASE 12

ANALYSIS PHASE 13

DESIGN PHASE 14

TESTING PHASE 14

DEPLOYMENT PHASE 15

VALIDATION PHASE 15

RELATED THEORIES 16

COMPARATIVE MATRIX 18

RESEARCH METHODOLOGY 22

RESEARCH DESIGN 22

RESEARCH ENVIRONMENT 24

RESPONDENTS 25

RESEARCH INSTRUMENT 26

SCORING PROCEDURE: ATTENDTRACK SYSTEM EVALUATION 30

DEFINITION OF TERMS 34

CHAPTER 2 37

PRESENTATION, DATA ANALYSIS AND INTERPRETATION RESPONDENTS PERCEPTION ON THE "ATTENDTRACK: A WEB-BASE EVENT ATTENDANCE AND EVALUATION SYSTEM" ACCORDING TO ISO/IEC 25010:2011 37

ATTENDTRACK SYSTEM EVALUATION 44

CHAPTER 3 49

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATION 49

SUMMARY OF FINDINGS 49

CONCLUSION 52

RECOMMENDATIONS 54

OUTPUT OF THE STUDY 55

RATIONALE 56

OBJECTIVES OF THE SYSTEM 57

TECHNICAL REQUIREMENTS OF ATTENDTRACK SYSTEM 58

PRODUCT BACKLOG 62

SYSTEM PROGRAM FLOW **65**

CONTEXT DIAGRAM 66

SUPER ADMIN USER FLOW 67

ADMIN USER FLOW 69

STUDENT USER FLOW 71

FUNCTIONAL DECOMPOSITION DIAGRAM 72

USE CASE DIAGRAM 73

ENTITY–RELATIONSHIP DIAGRAM 75

TEST RESULTS **77**

SOFTWARE REQUIREMENTS 79

HARDWARE REQUIREMENTS 81

RELEASE NOTES **84**

DEPLOYMENT PLAN **93**

DEPLOYMENT STEPS 93

DEPLOYMENT CONSIDERATIONS 99

USER GUIDE **101**

FUTURE RECOMMENDATIONS **108**

BIBLIOGRAPHY **111**

APPENDIX A 112

APPENDIX B 113

APPENDIX C 118

CURRICULUM VITAE **120**

LIST OF TABLES

Table	Contents	Page
1	Comparative Matrix	19
2	4-Point Agreement/Frequency/Effectiveness Scale	32
3	Mean Score Interpretation	32
4	Focus Areas & Survey Question Mapping	33
5	Inventory Management Gaps	32
6	Scoring, Analysis & Reporting Steps	34
7	Overall Age Distribution of Respondents	39
8	Overall Role in the Institution	40
9	Respondent’s Level of Experience with Digital Systems	41
10	Overall Respondent’s Department/Course	41
11	System Inefficiencies	42
12	Time Consumption Issues	42
13	Attendance Record Management Issues	43
14	Monitoring and Evaluation Issues	43
15	Technology and Automation Gaps	44
16	Centralized System Issues	44
17	Real-Time Monitoring Limitations	45
18	Integration and Evaluation Gaps	45
19	Functional Suitability	46
20	Performance Efficiency	47
21	Usability	48

22	Reliability	49
23	Security	50
24	Hardware and Software Requirements	60
25	Product Backlog	65

LIST OF FIGURES

Figure	Contents	Page
1	Iterative Methodology Framework	11
2	Flow of the study	23
3	Research Environment	24
4	Modified Three-Tier Architecture	60
5	Context Diagram	66
6	Super Admin User Flow	67
7	Admin User Flow	69
8	Student User Flow	70
9	Functional Decomposition Diagram	71
10	Use Case Diagram	73
11	Entity-Relationship Diagram (ERD)	75
12	UI/UX Flow	84

CHAPTER 1

THE PROBLEM AND ITS SCOPE

RATIONALE OF THE STUDY

Efficient event management and accurate attendance monitoring are essential components of institutional operations, particularly in educational environments where participation data is crucial for evaluation and decision-making. Manual attendance systems remain widely used despite being time-consuming and prone to human error, which adversely affects reporting accuracy and administrative efficiency (Bondoc, 2023). Research indicates that technology-driven attendance solutions especially those utilizing web platforms and QR code integrations significantly improve accuracy, transparency, and operational efficiency by automating data capture processes (Panjacharam& Harun, 2025; Pantilon, 2025). Web-based attendance systems also allow real-time data access and reduce workloads for staff and organizers (La Ode Muhammad Nurfansyah& Utami, 2025).

In many educational institutions, including Lapu-Lapu City College (LLCC), attendance tracking during events is still managed manually through paper registers or standalone spreadsheets, which leads to delays in record consolidation, errors, and increased workloads for faculty and administrative personnel. Studies on web-based QR code attendance systems report improvements in speed, accuracy, and user convenience compared with manual methods (Ikhsan & Helmina, 2022; Ghanvat et al., 2025). Furthermore, the application of web-based smart attendance systems with additional features such as SM S notifications and centralized reporting enhances monitoring and communication within school settings (Bondoc, 2023).

Despite the benefits of digital attendance tracking, most existing technologies are designed for classroom attendance or require multiple disconnected tools for event scheduling, attendance capture, and reporting. For instance, QR code-based lecture attendance systems focus primarily on classroom settings rather than comprehensive event management (Ikhsan & Helmina, 2022). Similarly, advanced QR code systems for school attendance streamline data capture but still lack integrated scheduling, analytics, and centralized administrative dashboards (Perwitasari et al., 2024). These limitations reveal a gap in the availability of a fully integrated web-based event management and attendance monitoring platform tailored to institutional event administration.

To address this gap, this study proposes the AttendTrack system, a web-based solution integrating event scheduling, participant registration, QR code attendance tracking, real-time dashboards, role-based access control, and automated reporting. This platform aims to reduce errors, enhance event coordination, improve data accuracy, and support real-time monitoring for administrators, organizers, and stakeholders. The design and evaluation of AttendTrack will be guided by user feedback and system usability standards to ensure effectiveness, reliability, security, and ease of use for its intended users.

The study's findings will contribute to the limited body of research on integrated web-based event and attendance systems in educational contexts. By demonstrating how the integration of QR technology, real-time analytics, and web platforms can streamline institutional processes, this research may serve as a reference for other colleges and universities seeking to modernize event management practices. Ultimately, AttendTrack supports institutional digital transformation, promotes efficient administrative workflows, and enhances data-driven decision-making in academic environments.

THE PROBLEM

Statement of the Problem

This study aimed to design, develop, and evaluate AttendTrack to address the challenges encountered by Lapu-Lapu City College (LLCC) in managing event attendance and participant evaluation through manual processes. These challenges included recording errors, delays in attendance tracking and report preparation, incomplete or lost records, encoding errors, increased workload, and difficulties in monitoring and evaluating event participation.

Specifically, this study seeks to answer the following questions:

1. What is the demographic profile of the respondents in terms of:
    a. Age
    b. Role in the institution (student, organizer, or administrator)
    c. Level of experience with digital or web-based attendance system

2. What problems do respondents experience with the existing manual event attendance recording in terms of:
    a. Inefficiencies (e.g., manual attendance recording and verification delays)
    b. Time consumption in tracking attendance and preparing reports
    c. Attendance record management (e.g., incomplete records, lost attendance sheets, and encoding errors)
    d. Monitoring and evaluating event participation

3. What are the underlying gaps that contribute to these problems, particularly in relation to:
    a. Lack of automation in attendance tracking and event evaluation
    b. Absence of a centralized system for event management and participation monitoring

c. Limited real-time visibility of attendance data during events
d. Inadequate integration of attendance monitoring and event evaluation tools

4. What extent do respondents assess the effectiveness of the AttendTrack System in addressing the problems encountered in the manual attendance process based on the effectiveness criterion of ISO/IEC 25010:2011:
    a. Functional Suitability
    b. Performance Efficiency
    c. Usability
    d. Reliability
    e. Security

5. What is the extent does AttendTrack address the identified problems of the manual system in terms of:
    a. Operational effectiveness in event management
    b. Process efficiency in attendance monitoring
    c. Accuracy and reliability of attendance records
    d. Improvement in event evaluation and feedback collection
    e. Overall user satisfaction

6. What recommendations can be proposed based on the findings to further improve the AttendTrack system and enhance event attendance monitoring and evaluation at Lapu-Lapu City College.

Objective of the Study

This study aims to design, develop, and evaluate AttendTrack: A Web-Based Event Attendance and Evaluation System for Lapu-Lapu City College as a digital solution for improving event management, attendance monitoring, and participant feedback collection while reducing the limitations of manual attendance processes.

Specific Objectives:

1. Develop and implement essential system functionalities including user authentication, role-based access control, event creation and scheduling, participant registration, QR code generation, attendance scanning, and automated report generation.
2. Enhance event monitoring and participation tracking through real-time attendance recording, centralized data storage, and system dashboards that allow administrators and organizers to monitor attendance data efficiently.
3. Integrate an event evaluation feature that enables participants to provide feedback and assessments of institutional events, allowing organizers to analyze event effectiveness and improve future activities.
4. Evaluate the system's quality and performance in terms of functional suitability, performance efficiency, usability, reliability, and security based on user feedback and system testing.
5. Determine the level of user satisfaction and system acceptance among students, organizers, and administrators when using AttendTrack for event attendance monitoring and evaluation.
6. Provide recommendations for further improvement of the system's features, usability, reporting capabilities, and scalability to support future institutional event management needs.

Significance of the Study

The development of AttendTrack: A Web-Based Event Attendance and Evaluation System is expected to provide several benefits to Lapu-Lapu City College and its stakeholders by improving the efficiency of event attendance monitoring and evaluation processes. The findings of this study may benefit the following:

Students

Students will benefit from a faster and more convenient way of recording their attendance during institutional events through QR code scanning. The system also allows them to provide feedback and evaluations about events, giving them an opportunity to share their experiences and suggestions for improvement.

Event Organizers

Event organizers will benefit from a more efficient system for managing events, tracking participant attendance, and collecting evaluation responses. The automated attendance monitoring and reporting features can help reduce manual work and allow organizers to focus more on planning and improving event activities.

Administrators

School administrators will benefit from centralized access to attendance records and evaluation reports. The system provides real-time monitoring and organized data that can support better decision-making, improve transparency, and enhance institutional event management.

Lapu-Lapu City College

The institution will benefit from a modernized and more reliable approach to event attendance monitoring and evaluation. The implementation of AttendTrack can help improve administrative efficiency, reduce errors in attendance documentation, and support data-driven evaluation of institutional events

Future Researchers.

This study may serve as a reference for future researchers who plan to develop or improve similar systems related to event management, attendance monitoring, and evaluation platforms in educational institutions.

Scope and Limitations of the Study

Scope of the Study

This study covers the design, development, and evaluation of AttendTrack, a web-based system developed to support event attendance monitoring and evaluation at Lapu-Lapu City College. The system aims to simplify event management processes by providing a digital platform that automates attendance recording and collects participant feedback for institutional events.

Key functionalities evaluated in this study include:

Event Management: Enables administrators and organizers to create, update, and manage event information such as event title, schedule, venue, and description.

User Registration and Access Control:

Allows students, organizers, and administrators to securely access the system based on assigned roles and permissions.

QR Code-Based Attendance Monitoring:

Generates unique QR codes for participants that can be scanned during events to automatically record attendance.

Real-Time Attendance Monitoring:

Allows organizers and administrators to track attendance instantly during events through a centralized dashboard.

Event Evaluation Module:

Provides an evaluation form that enables participants to submit feedback regarding the event, helping organizers assess the effectiveness and quality of activities.

Automated Reporting and Data Management: Generates attendance summaries, participation records, and evaluation reports that can be viewed or exported for documentation and analysis.

The AttendTrack system is implemented as a web-based platform that can be accessed through modern web browsers using devices such as computers or smartphones with internet connectivity. The system is intended specifically for institutional events conducted within Lapu-Lapu City College, particularly those organized by the College of Technology and other authorized departments.

Limitations of the Study

While the AttendTrack system offers significant improvements in event attendance monitoring and evaluation, the study is subject to several limitations:

Internet Dependency for System Use:

The functionality of the system relies heavily on stable internet access. Any disruption in connectivity may affect real-time attendance recording and system responsiveness during events.

Institution-Specific Implementation:

The system is developed and tested specifically for Lapu-Lapu City College. Its design and configuration may not be directly applicable to other institutions without further customization or modification.

Focus on Event-Based Attendance Only:

The system is intended primarily for managing attendance during institutional events and does not extend to regular classroom attendance or academic tracking.

Limited Advanced Features:

Certain advanced technologies such as biometric authentication, offline attendance capability, and mobile application integration are not included in the current version of the system.

Restricted User Scope:

The evaluation of the system is based mainly on student respondents, which may limit the diversity of feedback from other potential users such as administrators and event organizers.

Performance Testing Constraints:

The system was tested within a controlled environment and moderate number of users. Its performance under large-scale usage or high concurrent access was not extensively evaluated.

Theoretical Background

Software Development Project Framework

Model The proponents adopted an Iterative Methodology Framework as the foundation for designing, developing, and implementing the AttendTrack: A Web-Based Event Attendance and Evaluation System in in Lapu-Lapu City College. As illustrated in the provided diagram, the framework follows a cyclical process where each phase planning, analysis, design, implementation, testing, review, and validation continuously interacts with one another. This iterative approach emphasizes repeated refinement, ensuring that outputs from each stage are evaluated and improved before proceeding. By following this structure, the system development remains aligned with user requirements while maintaining flexibility to adapt to changes and emerging needs.

The iterative framework supports continuous development and evaluation, allowing the system to evolve through multiple cycles rather than a single linear process. Based on the diagram, the review phase plays a crucial role in feeding insights back into earlier stages, enabling enhancements in system features such as QR code-based attendance tracking, real-time monitoring, and notification integration. This process ensures that errors are identified early, usability is improved, and system performance is optimized. Additionally, the validation phase confirms that the system meets key standards, including functionality, reliability, security, and usability, before moving forward or finalizing outputs. By implementing this methodology, the proponents aimed to develop a responsive, accurate, and user-centered attendance monitoring system. The iterative development process enables continuous feedback and system improvements, reducing errors while increasing efficiency. It also allows the system to adapt to the needs of student attendance management and administrative monitoring, resulting in a reliable, effective, and user-friendly solution.

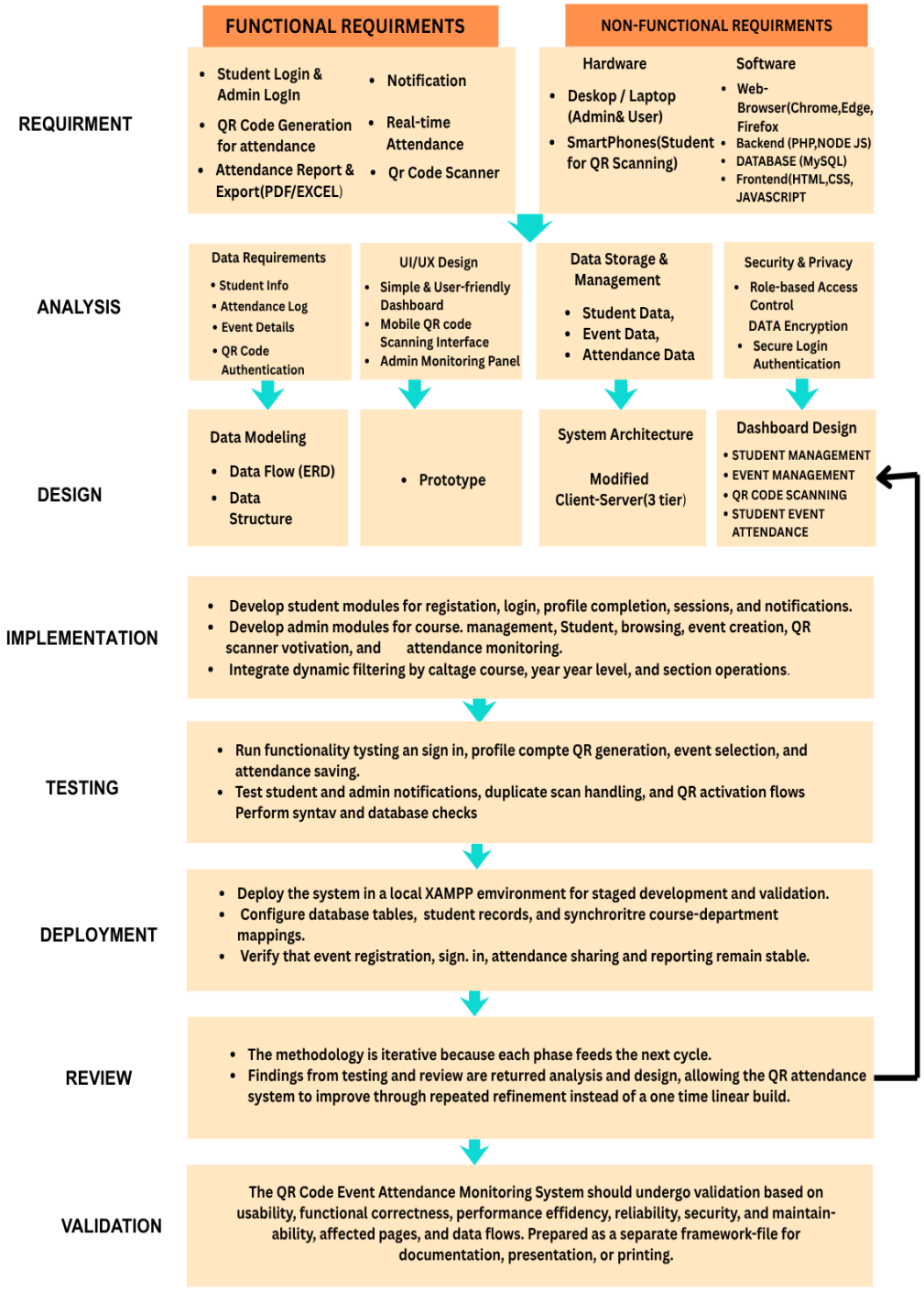


Figure 1: *Iterative Methodology Framework*

Iterative Development Framework

This study follows an Iterative Development Framework in developing the AttendTrack system. The process includes the phases of Requirements, Analysis, Design, Coding, Testing, Implementation, and Review. Functional requirements such as user registration, QR code generation and scanning, attendance reports, notifications, user management, and dashboard analytics are defined alongside non-functional requirements covering hardware, software, and platform specifications. Each phase is continuously refined through analysis, system design, development, and ISO 25010-based testing, including User Acceptance Testing (UAT). After deployment, the system is evaluated for performance, accuracy, and reliability to ensure continuous improvement and adaptability.

Requirements Phase

The requirements phase focuses on identifying and defining the functional and non-functional requirements of the proposed QR Code Event Attendance Monitoring System. This phase establishes the foundation of the system by determining the necessary features, system capabilities, and constraints based on user needs and operational objectives.

Here, requirements are translated into precise technical and user specifications:

1. Functional requirements specify the core features of the system, including student and administrator login, QR code generation for attendance, QR code scanning, real-time attendance recording, notification mechanisms, and the generation of attendance reports in formats such as PDF and Excel. These features ensure that the system effectively supports attendance monitoring and event management processes.

2. Non-functional requirements, on the other hand, define the system's operational environment and performance expectations. These include hardware requirements such as desktops or laptops for administrators and smartphones for students to enable QR code scanning. Software requirements include web browsers (e.g., Chrome and Edge), backend technologies such as PHP and Node.js, a MySQL database, and frontend technologies including HTML, CSS, and JavaScript. Additionally, system performance, usability, reliability, and responsiveness are considered to ensure efficient and user-friendly operation.

Analysis Phase

During the analysis phase, system requirements are translated into comprehensive technical and user-oriented specifications. Data requirements are identified to support the management of student information, attendance logs, event details, and QR code authentication, ensuring accurate and efficient tracking of attendance.

Furthermore, user interface and user experience (UI/UX) design considerations focus on developing a simple and intuitive dashboard, a mobile-compatible QR code scanning interface, and an administrative monitoring panel to enhance usability and accessibility across platforms.

Design Phase

During the design phase, the system architecture and components are structured based on the analyzed requirements:

Data modeling is conducted through the development of Entity-Relationship Diagrams (ERD) to define relationships among students, fees, transactions, and notifications.

A prototype of the user interface is developed to visualize workflows and gather early feedback on usability.

A modified three-tier client-server architecture is designed, consisting of presentation, application, data, and integration layer. Dashboard components are designed to provide role-based modules for fee management, transaction monitoring, and notifications.

The output of this phase is a development-ready system blueprint for implementation.

Implementation PhaseWith the design finalized, the focus shifts to system development and configuration: Core modules are developed for student profiles, fee management, payment processing, and transaction history. The system interface is built using Tailwind CSS and Bootstrap for responsive design, while PHP and Laravel are used for backend logic and system integration. User authentication features such as login, registration, and role-based access control are implemented. SMS gateway integration is developed to support automated reminders and notifications for users.

Testing Phase

The testing phase ensures that all system functionalities meet quality and performance standards: User testing is conducted with administrators, staff, and users to evaluate usability and system performance. Functional testing is performed on payment processing, dashboard operations, authentication, and SMS notification features. Error handling is tested, including invalid login attempts, failed transactions, and data inconsistencies. Database integrity and system reliability are validated following ISO/IEC 25010:2011 standards.

Deployment Phase

During the deployment phase, the system is prepared for real-world use:

The system is initially deployed in a local server environment for controlled testing and configuration. After validation, it is transitioned to a production or cloud-based hosting environment. Database setup is completed, including tables for users, fees, transactions, and notifications. System functionality is verified to ensure smooth operation across different devices and browsers.

Review Phase

The review phase focuses on system evaluation and continuous improvement:

User feedback is collected and analyzed to identify usability and performance issues. System bugs and inconsistencies are documented and prioritized for correction.

Enhancements are implemented based on user needs and system performance evaluation. The process follows an iterative cycle to continuously improve system efficiency and usability.

Validation Phase

The validation phase ensures the system meets all defined requirements and standards:

The system is evaluated based on usability, functionality, performance efficiency, reliability, security, and maintainability. Comprehensive testing confirms that all core features operate as intended. The system is verified to support automated fee processing, transaction tracking, and notification services. Final validation results are documented for system approval and academic submission.

Related Theories

Web-based attendance systems have become increasingly essential in educational institutions, particularly in addressing the operational challenges identified at Lapu-Lapu City College (LLCC). As highlighted in the rationale of this study, manual attendance methods such as paper-based recording and spreadsheet encoding remain widely used but are often inefficient, time-consuming, and prone to human error, resulting in inaccurate records and delays in report generation. Studies have shown that web-based attendance systems utilizing QR code technology significantly improve data accuracy, automate attendance capture, and enable real-time monitoring of participation (Bondoc, 2023; Ikhsan & Helmina, 2022). These capabilities directly support the objective of the AttendTrack system to enhance process efficiency, accuracy, and real-time visibility of attendance data during institutional events.

Furthermore, the integration of web and mobile technologies enhances accessibility and usability, allowing users to conveniently access attendance data across devices. This supports centralized data management and improves communication among stakeholders, which addresses the identified problem of lacking a unified platform for attendance monitoring (La Ode Muhammad Nurfansyah& Utami, 2025). This technological integration is reflected in AttendTrack's design as a web-based system accessible via modern browsers, enabling students, organizers, and administrators to interact with the platform efficiently.

In terms of event management, modern web-based systems incorporate functionalities such as event scheduling, participant registration, attendance tracking, and automated reporting, which are also core features of AttendTrack. Research indicates that systems with role-based access control enhance security and allow users to interact based on their assigned responsibilities (Panjacharam& Harun, 2025). This supports the system design of AttendTrack, where administrators, organizers, and students have

specific roles and permissions. Additionally, QR code-based attendance systems provide a faster and more reliable method of verifying participants compared to manual processes, minimizing issues such as proxy attendance and delayed verification (Ghanvat et al., 2025).

However, despite these advancements, many existing systems are limited in functionality, as they primarily focus on attendance tracking without integrating evaluation and feedback mechanisms (Ikhsan & Helmina, 2022; Perwitasari et al., 2024). This limitation creates a gap in assessing event effectiveness and participant satisfaction. AttendTrack addresses this gap by incorporating an event evaluation module that allows participants to provide feedback, aligning with the study's objective of improving event assessment and institutional decision-making.

Moreover, evaluation and monitoring systems play a crucial role in enhancing system effectiveness and supporting continuous improvement. The integration of evaluation tools enables institutions to collect feedback, analyze participation trends, and make data-driven decisions. QR code-based attendance systems have been found to improve monitoring efficiency while providing accurate data for decision-making processes (Pantilon, 2025). In line with this, AttendTrack is evaluated based on software quality standards such as ISO/IEC 25010, which emphasize key attributes including functionality, usability, reliability, performance efficiency, and security. These criteria are consistent with the study's research objectives and ensure that the system meets both user expectations and institutional requirements.

Overall, the studies support the development of AttendTrack as an integrated solution for improving attendance and event management at LLCC through QR technology, real-time monitoring, and data-driven features (Bondoc, 2023; Panjacharam & Harun, 2025; Pantilon, 2025).

Comparative Matrix

***Table 1**: Comparative Matrix*

Features	SYSTEMS				
	AttendTrack: A Web-Based Event Attendance and Evaluation System in Lapu-Lapu City College (Proposed System)	Web-based smart attendance monitoring system using QR code and GSM module for SMS notification	Design of a Web-Based Lecturer Attendance Information System Using QR Codes at Muhammadiyah University of Jambi	SP Mega Mariharta Staff Management System and Dynamic QR Code Attendance System with Role Based Access Control	Digitized Attendance Recording System Using QR Technology
Web-Based Platform	✓	✓	✓	✓	✓
QR Code Attendance	✓	✓	✓	✓	✓
Real-Time Monitoring	✓	✓	✗ (No live dashboard)	✓	✗ (No real-time tracking)
Event Management	✓	✗ (Attendance only)	✗ (Classroom only)	✗ (No event module)	✗ (Tracking only)
Participant Registration	✓	✗ (No registration feature)	✗ (Pre-registered users only)	✗ (No registration module)	✗ (No user registration)
Role-Based Access Control	✓	✗ (Single user role)	✗ (No role management)	✓	✗ (No role control)
Automated Reporting	✓	✓	✓	✓	✓
Notification System	✓	✓	✗ (No notification feature)	✗ (No alerts/notifications)	✗ (No notification system)
Evaluation/Feedback Module	✓	✗ (No evaluation feature)	✗ (No feedback system)	✗ (No evaluation module)	✗ (No evaluation feature)
Data Analytics Dashboard	✓	✗ (No analytics dashboard)	✗ (Basic records only)	✗ (No dashboard analytics)	✗ (No data visualization)
Security Features	✓	✓	✓	✓	✓

Legend: ✔ – Feature is available. *✗ – The feature is not available*

Related System

Several existing systems were examined to identify their strengths, weaknesses, and functional limitations, which served as a basis for the development of AttendTrack: A Web-Based Event Attendance and Evaluation System in Lapu-Lapu City College. These systems provided insights into current practices in event management, attendance tracking, and evaluation, highlighting the need for a more integrated and centralized solution.

AttendTrack: A Web-Based Event Attendance and Evaluation System in Lapu-Lapu City College

(Proposed System)

Unlike existing systems, AttendTrack is a centralized web-based platform developed to manage institutional events, monitor attendance, and evaluate participant engagement within Lapu-Lapu City College. The system integrates key functionalities such as QR code-based attendance scanning, real-time monitoring dashboards, role-based access for administrators, organizers, and students, and automated report generation. In addition, it includes an evaluation module that allows participants to provide feedback after events. These combined features address the limitations of traditional attendance systems by unifying attendance tracking and event evaluation into a single platform, thereby improving operational efficiency, data accuracy, and decision-making processes.

Web-based smart attendance monitoring system using QR code and GSM module for SMS notification

This system is a web-based attendance solution developed specifically for classroom environments, where QR codes are used to automate attendance recording and store data digitally. It improves efficiency and reduces human errors associated with manual attendance processes. Despite these benefits, the system is limited to academic settings and does not support event-based attendance, multi-role user access, real-time monitoring dashboards, or evaluation features, which are essential components integrated into the AttendTrack system.

Design of a web-based lecturer attendance information system using QR codes.

This system is a web-based attendance solution developed specifically for classroom environments, where QR codes are used to automate attendance recording and store data digitally. It improves efficiency and reduces human errors associated with manual attendance processes. Despite these benefits, the system is limited to academic settings and does not support event-based attendance, multi-role user access, real-time monitoring dashboards, or evaluation features, which are essential components integrated into the AttendTrack system.

Dynamic QR code attendance system with role-based access control.

This system utilizes dynamic QR codes and role-based access control to provide a secure and flexible attendance monitoring solution. It allows multiple users to interact with the system based on assigned roles, enhancing system security and organization. While it effectively improves attendance tracking and access management, the system lacks integrated evaluation features and does not offer a complete event management solution. Compared to AttendTrack, it does not include functionalities such as participant feedback collection, centralized event scheduling, and comprehensive reporting.

Digitized Attendance Recording System Using QR Technology

This study presents a digitized attendance system that utilizes QR code technology to streamline the process of recording attendance. The system enhances speed and accuracy in verifying participants and minimizes common issues such as duplicate entries and proxy attendance. It provides a more efficient alternative to manual attendance methods. However, the system focuses primarily on attendance recording and does not include advanced features such as event scheduling, real-time dashboards, or evaluation modules, making it less comprehensive compared to AttendTrack.

RESEARCH METHODOLOGY

Research Design

This study adopteda descriptive quantitative research design was utilized to evaluate the effectiveness and usability of AttendTrack: A Web-Based Event and Attendance Monitoring and Evaluation System. This approach was appropriate as it enabled the researchers to systematically measure user experiences, system performance, and overall functionality within an actual event management setting. Quantitative data were collected through structured survey questionnaires and system-generated evaluation metrics to assess key factors such as ease of use, efficiency of attendance tracking, accuracy of recorded data, timeliness of monitoring processes, and overall user satisfaction. This design is consistent with established software quality evaluation standards and information systems assessment models, which emphasize objective and measurable evaluation of system performance (ISO/IEC, 2011; Kitchenham et al., 2017).

Additionally, the use of survey-based instruments aligns with recent studies that assess user acceptance and effectiveness of digital systems in educational and organizational environments (Scherer et al., 2019; Al-Emran et al., 2020). The collected data were analyzed using appropriate statistical techniques to determine how effectively AttendTrack addressed common issues in traditional event management, such as manual attendance recording, delayed reporting, lack of real-time tracking, and limited communication among stakeholders. The results provided an evidence-based evaluation of the system’s usability, performance, and effectiveness in improving event attendance monitoring and evaluation processes.

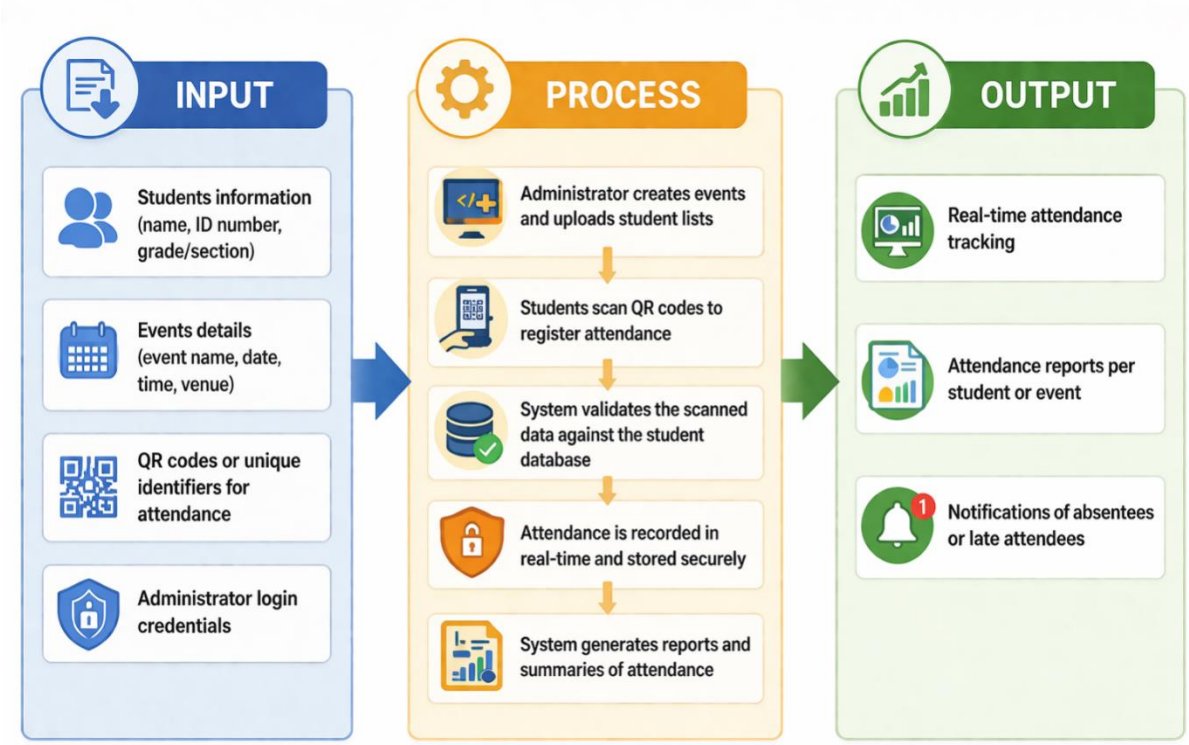


***Figure 2**: Flow of the study*

Figure 2 shows the study's flow using an Input–Process–Output framework, where system requirements for the QR-based Attendance Monitoring System are gathered through analysis of student information, event details, QR code identification methods, and administrator login requirements, then examined to define the system specifications. The system is designed and developed through the implementation of modules for event creation, QR code scanning, database validation, real-time attendance recording, and automated report generation, which are integrated and refined through testing and user feedback. These activities result in a fully implemented attendance monitoring system that enables real-time tracking of student attendance, generates attendance reports per student or event, and sends notifications for absences or late entries, ensuring accurate, secure, and efficient attendance management.

Research Environment

This study was conducted at Lapu-Lapu City College (LLCC), where the AttendTrack system will be implemented and tested. The environment includes classrooms, computer laboratories, and event venues within the school, providing a realistic setting for monitoring student attendance and participation. LLCC is equipped with modern technological facilities, including internet enabled devices and computer labs, which support the use of web-based systems.



***Figure 3**: Research Environment*

Respondents

The respondents of this study consist of students from Lapu-Lapu City College who will utilize AttendTrack: A Web-Based Event Attendance and Evaluation System, as the system is specifically designed to help them monitor and record attendance during school events. Their feedback will be used to assess the system's usability, functionality, and overall effectiveness. Participants will be selected from different year levels and academic programs to ensure a diverse range of perspectives. A total of fifty (50) respondents will be involved in the study, including student users, as well as designated roles such as the Super Admin and Admin. The Super Admin and Admin will evaluate the system's administrative and management features, while the student users will focus on its accessibility and ease of use. Their combined responses will help determine the overall performance of AttendTrack, identify any issues encountered, and provide recommendations for further system improvement.

Sampling Technique

This study will utilize purposive sampling to select respondents who will evaluate the AttendTrack system. This non-probability sampling method allows the researchers to intentionally choose participants who have direct experience with event attendance monitoring and can provide relevant feedback for the study. The respondents will include selected students, faculty members, and event organizers from Lapu-Lapu City College who are involved in institutional events where attendance tracking is required. These participants are considered suitable because they regularly interact with attendance recording processes and can provide valuable insights regarding the usability, effectiveness, and reliability of the proposed web-based attendance and evaluation system. The researchers will coordinate with the college administration and event coordinators to identify qualified respondents and ensure that sufficient data is gathered.

Research Instrument

To collect the necessary data for evaluating the proposed AttendTrack system, the researchers will utilize a structured survey questionnaire as the primary research instrument. The questionnaire will be created using Google Forms to allow convenient distribution and efficient collection of responses from the selected participants. It will be designed based on the objectives and research questions of the study to ensure that all important aspects of the system are properly assessed.

The research instrument will include the following components:

Demographic Information

This section will gather basic information about the respondents, including their role in the institution (student, faculty member, or event organizer) and their familiarity with digital attendance systems.

Assessment of the Current Attendance Monitoring Method

This section will identify the respondents' experiences with the existing manual attendance recording process, including issues related to time consumption, record accuracy, and administrative workload.

Evaluation of the AttendTrack System

This section will measure the respondents' perceptions of the system using a Likert scale, focusing on software quality attributes such as functionality, usability, efficiency, reliability, and security.

System Impact Assessment

This part will evaluate how the AttendTrack system improves attendance monitoring, accuracy of records, reporting efficiency, and user convenience.

Open-Ended Feedback

This section will allow respondents to provide suggestions and recommendations for further improvement of the AttendTrack system.

Data Gathering

The data gathering for this study will follow a systematic process to ensure reliable and valid information, covering the preparation, distribution, and collection of data for evaluating the AttendTrack system.

Preparation Phase

In the preparation phase, researchers will create the research instruments, including a structured Google Forms questionnaire and system demonstration materials, to collect data on current manual attendance practices and the effectiveness of the AttendTrack system.

Approval and Consent

Ethical approval and informed consent were obtained from the selected schools and participants to ensure compliance with research ethics standards. Participation was voluntary, and respondents were assured of confidentiality.

Identification of Respondents

The researchers will coordinate with the college administration to select participants for the study, including students, faculty members, and event organizers who are actively involved in college events. These respondents are chosen because of their experience with attendance monitoring, ensuring they can provide reliable and meaningful feedback on the effectiveness and usability of the AttendTrack system.

Distribution of Instruments

The survey questionnaire will be distributed to the selected respondents online through Google Forms, allowing them to complete it conveniently on their personal devices. Before answering, participants will receive a demonstration of the AttendTrack system, including its QR-based attendance recording and evaluation features, ensuring they understand the system and can provide informed and accurate feedback on its performance.

System Logs and Follow-up Interviews

To supplement the survey data, the researchers will review records generated by the AttendTrack platform, including actual attendance activities and system interactions during events. Additionally, informal follow-up discussions with selected respondents may be conducted to gain deeper insights into user

Data Compilation, Security, and Validation

All data collected from respondents will be compiled and organized for analysis, with responses from Google Forms exported to spreadsheets for accurate processing. To ensure confidentiality, all information will be securely stored and accessible only to authorized researchers, and personal identifiers will not be disclosed.

Before analysis, data will be validated for completeness and accuracy, excluding unrealiable responses to ensure a credible evaluation of the AttendTrack System

Statistical Treatment of Data

Descriptive statistical tools will be used to organize, present, analyze, and interpret data collected from respondents through survey questionnaires in evaluating the AttendTrack system, particularly in terms of its effectiveness, usability, and overall performance.

1. Weighted Mean will be used to determine the level of respondents' evaluation of the AttendTrack system in terms of software quality characteristics such as functional suitability, performance efficiency, usability, reliability, and security, as well as its effectiveness in improving event attendance monitoring and evaluation.

   Formula:

   $$\bar{x} = \frac{\sum fx}{N}$$

   Where:

   $\bar{x}$ is the weighted mean

   f is the frequency

   x are the weights assigned

   N is the total number of respondents

2. Standard Deviation will be used to measure the degree of variability or dispersion of the responses. This will indicate how consistent the respondents' evaluations are regarding the system's usability, reliability, and performance.

   Formula:

   $$s = \sqrt{\frac{\sum(X - \bar{x})^2}{n - 1}}$$

   Where:

   s is the sample standard deviation

   $\bar{x}$ is the sample mean

   X is each individual sample value

   n is the total number of samples

3. Frequency and Percentage Distribution will be used to present the respondents' demographic profile and categorical responses, providing a clear overview of their characteristics and response trends.

   Formula:

$$\% = (\frac{f}{N}) \times 100$$

   Where:

   f is the frequency of a particular response

   N is the total number of respondents

Scoring Procedure: AttendTrack System Evaluation

The scoring procedure was applied to assess the effectiveness, usability, and overall performance of AttendTrack: A Web-Based Event Attendance and Evaluation System, based on the data gathered from the structured survey questionnaire.

Responses from Section D (Evaluation of the AttendTrack System) and Section E (Effectiveness of the AttendTrack System) were evaluated using a four-point Likert scale, where:

1. Likert Scales Used in the Questionnaire

Your questionnaire uses multiple 4-point Likert scales across sections, depending on focus area. These scales are standardized for scoring purposes

Each questionnaire item was given a numerical value based on the respondent's selected answer. The weighted mean was then calculated for each indicator and category to assess the overall evaluation of the system.

***Table 2**: 4-Point Agreement/Frequency/Effectiveness Scale*

Scale Type	Verbal Interpretation
4	Strongly Agree
3	Agree
2	Disagree
1	Strongly Disagree

2. Interpretation of Mean Scores

The mean scores showed how effectively the AttendTrack system addressed user needs. Higher values indicated better system performance, while lower values pointed to aspects that needed improvement. This enabled the researchers to clearly classify the systems effectiveness and usability based on participants responses.

Table 3: Mean Score Interpretation

Mean Score Range	Interpretation
3.25 – 4.00	Strongly Agree
2.50 – 3.24	Agree
1.75 – 2.49	Disagree
1.00 – 1.74	Strongly Disagree

3. Focused Scoring Areas Mapped to Survey Questions

The questionnaire has five main evaluation sections (A-E). Each section is mapped to specific research questions and Scoring metrics.

Table 4: Focus Areas & Survey Question Mapping

Section	Focus Area / Sub-Area	Survey Questions / Metrics
Section A	Demographics	Age, Role (Student, Instructor, Administrator), Experience with Digital Attendance Monitoring Systems (Q1–Q3)
Section B	Manual Attendance Monitoring Problems	Manual attendance recording errors, delays in attendance verification, time-consuming tracking, lost or incomplete records, delayed report generation, and communication issues
Section C	Root Causes & System Gaps	Lack of automation, absence of real-time attendance tracking, no centralized database, limited accessibility, poor system integration, inadequate security, inefficient record management
Section D	AttendTrack System Evaluation	1. Functionality, 2. Usability, 3. Reliability, 4. Accuracy, 5
Section E	AttendTrack System Effectiveness	1. Operational Effectiveness, 2. Process Efficiency, 3. Accuracy of Attendance Records, 4. Communication Improvement, 5. User Satisfaction

4. Scoring and Priority Ranking for System Improvements

After evaluating quantitative responses, areas needing improvement are prioritized using the following scale.

***Table 5**: System Improvement Priority Ranking*

Priority Level	Numerical Value	Description
Highest	5	Critical; implement immediately for core functionality
High	4	Essential; major user pain point affecting adoption
Medium	3	Beneficial; improves efficiency and satisfaction
Low	2	Nice-to-have; minor refinements
Lowest	1	Optional; low impact on objectives

5. Steps for Scoring, Analysis, and Reporting

Responses were coded numerically, and mean scores were calculated for each focus area. Differences between respondent groups were analyzed, and open-ended feedback was reviewed thematically to identify strengths, weaknesses, and areas for improvement of the AttendTrack system.

***Table 6**: Scoring, Analysis & Reporting Steps*

Steps	Description
Response Coding	Assign numerical values (1–4) to all Section D & E items based on the Likert scale.
Mean Calculation	Compute mean scores per focus area (Functionality, Usability, Reliability, Accuracy, Security, Operational Effectiveness, Process Efficiency, etc.).
Interpretation	Compare mean scores with the interpretation table to determine system performance levels.
Group Analysis	Compare scores among students, coordinators, administrators to identify differences in perception.
Aggregated Indices	Compute overall indices for AttendTrack System Effectiveness and Event Attendance Improvement using combined scores.

Definition of Terms

AttendTrack System

Conceptual: A web-based platform designed to manage and monitor event attendance and evaluation processes in an institution.

Operational: The AttendTrack: A Web-Based Event Attendance and Evaluation System developed in this study, which includes features such as QR code attendance scanning, event management, dashboards, and automated reporting.

Event Management

Conceptual: The process of planning, organizing, and executing events efficiently.

Operational: The feature in the system that allows administrators and organizers to create, schedule, and manage institutional events.

Attendance Monitoring

Conceptual: The process of tracking and recording participants' presence.

Operational: The system function that records attendance through QR code scanning and displays participation data in real time.

QR Code

Conceptual: A machine-readable code used to store and retrieve information.

Operational: The code generated by the system for each participant, which is scanned to record attendance automatically.

User Authentication

Conceptual: The process of verifying the identity of a user before granting access.

Operational: The login mechanism in the system that requires valid credentials such as username and password.

Role-Based Access Control

Conceptual: A method of restricting system access based on user roles.

Operational: The feature that assigns permissions to users such as administrators, organizers, and students.

Dashboard

Conceptual: A centralized interface displaying summarized system information.

Operational: The section of the system where users view attendance summaries, reports, and system activities.

Event Evaluation

Conceptual: The process of assessing the effectiveness of an event.

Operational: The feature that allows participants to submit feedback and ratings after attending events.

Automated Reporting

Conceptual: The generation of reports without manual processing.

Operational: The system feature that produces attendance summaries and evaluation reports automatically.

Real-Time Monitoring

Conceptual: The ability to track activities instantly as they occur.

Operational: The system capability that updates attendance data immediately after QR code scanning.

Centralized Database

Conceptual: A unified storage system for managing data.

Operational: The database used to store user information, attendance records, and event data.

Data Accuracy

Conceptual: The correctness and reliability of data.

Operational: The system's ability to record and maintain error-free attendance and evaluation data.

System Usability

Conceptual: The ease with which users can interact with a system.

Operational: Measured through user feedback on navigation, accessibility, and overall experience of the system.

Administrative Efficiency

Conceptual: The improvement of administrative tasks through technology.

Operational: Evaluated by comparing manual attendance processes with the automated functions of the system.

CHAPTER 2

PRESENTATION, DATA ANALYSIS AND INTERPRETATION RESPONDENTS PERCEPTION ON THE “Attendtrack: A Web-Base Event Attendance And Evaluation System” ACCORDING TO ISO/IEC 25010:2011

This section presents and interprets the data gathered during the evaluation of the AttendTrack: A Web-Based Event Attendance and Evaluation System**.** It includes the respondents' demographic profile, the challenges encountered in the existing manual event attendance process, the identified technological gaps, and the assessment of the proposed system based on the ISO/IEC 25010:2011 Software Quality Model. Data were collected from 50 respondents**,** composed of students, event organizers, and administrators of Lapu-Lapu City College, using a structured questionnaire. The responses were analyzed through frequency, percentage, weighted mean, and standard deviation to determine the system's quality in terms of functional suitability,performance efficiency, usability, reliability, and security**,** as well as its effectiveness in improving event attendance monitoring and evaluation.

Demographic Profile of Respondents: Age Distribution by Role

This section presents the respondents’ age distribution by role in relation to the AttendTrack system, illustrating how different age groups are represented among users such as students, administrators, and event facilitators. It highlights observable patterns in age variation across roles and identifies whether specific roles are primarily composed of younger or older individuals. This information provides valuable insight into how age may influence users’ familiarity with QR-based attendance systems, adaptability to digital event management platforms, and overall interaction with the AttendTrack system, helping assess usability and accessibility across diverse user groups.

***Table 7**: Overall Age Distribution of Respondents*

AGE GROUP	FREQUENCY	PERCENTAGE
18–20 years old	24	48%
21–23 years old	18	36%
24–26 years old	7	14%
27 years old & above	1	2%
Total	50	100%

Table 7 presents the age distribution of the respondents who participated in the study. The data show that nearly half of the respondents (48%) are 18–20 years old, making this the largest age group. This is followed by respondents aged 21–23 years old, who make up 36% of the total. Meanwhile, 14% of the respondents are 24–26 years old, while only 2% are 27 years old and above. These results indicate that most of the participants are young adults, which is expected since the study was conducted in a college setting.

***Table 8**: Overall Role in the Institution*

ROLE IN INSTITUTION	FREQUENCY	PERCENTAGE
Student	40	80%
Event Organizer	5	10%
Administrator	5	10%
Total	50	100%

Table 8 presents the respondents based on their roles in the institution. Most of the respondents were students (80%), while event organizers and administrators each made up 10%. This shows that the study mainly gathered feedback from students, who are the primary users of AttendTrack, while the event organizers and administrators provided additional perspectives on event management and attendance monitoring.

***Table 9**: Respondent's Level of Experience with Digital Systems*

EXPERIENCE LEVEL	FREQUENCY	PERCENTAGE
No experience	19	38%
Limited experience	15	30%
Moderate experience	8	38%
Extensive experience	8	16%
Total	50	100%

Table 9 presents the respondents' level of experience with digital or web-based attendance systems. The results show that 19 respondents (38%) had no experience, while 15 respondents (30%) had limited experience. Meanwhile, 8 respondents (16%) had moderate experience, and another 8 respondents (16%) had extensive experience. The results indicate that a considerable number of respondents had limited or no prior experience with digital attendance systems, highlighting the importance of making AttendTrack user-friendly, accessible, and easy to understand.

***Table 10**: Overall Respondent's Department/Course*

DEPARTMENT/COURSE	FREQUENCY	PERCENTAGE
College of Technology	16	32%
College of Education	25	50%
College of Hospitality & Tourism Management	9	18%
Total	50	100%

Table 10 presents the respondents according to their department or course. The College of Education had the highest number of respondents at 50%, followed by the College of Technology at 32% and the College of Hospitality and Tourism Management at 18%. The distribution shows that respondents came from different academic backgrounds, providing varied perspectives on the AttendTrack system and its usability for event attendance and evaluation.

Respondents' Perception

Problems in the Current Manual Event Attendance System

***Table 11**: System Inefficiencies*

PROBLEM	FREQUENCY	PERCENTAGE
Manual attendance recording causes delays	39	78%
Slow verification of participant attendance	25	50%
Difficulty consolidating attendance data	18	36%
Increased administrative workload	18	36%
Total Respondents	50	100%

Manual recording delays (78%) are the most critical inefficiency. Half of respondents also noted slow verification, while over one-third reported consolidation difficulties and increased workload. This shows inefficiency is systemic, affecting both speed and accuracy.

***Table 12**: Time Consumption Issues*

PROBLEM	FREQUENCY	PERCENTAGE
Attendance tracking is time-consuming	33	66%
Slow preparation of attendance reports	19	38%
Delays in updating participant records	24	48%
Manual monitoring takes too long	25	50%
Total Respondents	50	100%

Two-thirds (66%) agreed attendance tracking is time-consuming. Half cited manual monitoring as too long, while nearly half reported delays in updating records. This confirms time inefficiency is a major weakness of the manual system.

Table 13: *Attendance Record Management Issues*

PROBLEM	FREQUENCY	PERCENTAGE
Incomplete attendance records	32	64%
Lost attendance sheets	26	52%
Encoding errors	30	60%
Duplicate attendance entries	16	32%
Total Respondents	50	100%

Incomplete records (64%) and encoding errors (60%) are the most common issues. More than half also experienced lost sheets. These findings highlight the unreliability of manual documentation, which compromises data integrity and accountability.

Table 14*: Monitoring and Evaluation Issues*

PROBLEM	FREQUENCY	PERCENTAGE
Difficulty tracking participant involvement	28	56%
Lack of real-time attendance monitoring	32	64%
Difficulty evaluating event participation	21	42%
Limited access to attendance data	24	48%
Total Respondents	50	100%

The most pressing monitoring issue was lack of real-time visibility (64%). Over half also reported difficulty tracking involvement, while nearly half cited limited access to data. These results confirm manual systems fail to provide timely and comprehensive evaluation.

Root Causes and System Gaps

Table 15: *Technology and Automation Gaps*

GAP	FREQUENCY	PERCENTAGE
Lack of automation	32	64%
Manual data consolidation	32	64%
No QR code system	22	44%
No automated reports	14	28%
Total Respondents	50	100%

Respondents identified lack of automation and manual data consolidation (both 64%) as the most critical gaps. Nearly half noted the absence of a QR code system, while over a quarter highlighted the lack of automated reports. These results confirm that the manual system is technologically outdated and unable to support streamlined processes.

Table 16: *Centralized System Issues*

ISSUE	FREQUENCY	PERCENTAGE
No centralized database	29	58%
No unified system	24	48%
Data stored in multiple locations	20	40%
Limited access to past data	24	48%
Total Respondents	50	100%

The absence of a centralized database (58%) was the most common issue, followed by lack of a unified system and limited access to past data (both 48%). Forty percent also reported fragmented data storage. This indicates that the manual system fails to provide a single source of truth, making retrieval and management inefficient.

Table 17: *Real-Time Monitoring Limitations*

LIMITATION	FREQUENCY	PERCENTAGE
No real-time visibility	27	54%
Delayed updates	25	50%
Cannot track participation live	25	50%
No live reporting	16	32%
Total Respondents	50	100%

More than half of respondents (54%) reported lack of real-time visibility. Half also cited delayed updates and inability to track participation live. Nearly one-third noted absence of live reporting. These findings highlight that the manual system cannot provide timely insights, limiting responsiveness during events.

Table 18: *Integration and Evaluation Gaps*

GAP	FREQUENCY	PERCENTAGE
Not linked to evaluation tools	29	58%
No feedback collection	26	52%
Separate systems used	27	54%
Limited analytics	21	42%
Total Respondents	50	100%

The most common integration gap was lack of linkage to evaluation tools (58%). Over half also reported absence of feedback collection and reliance on separate systems. Limited analytics (42%) further emphasized the system's inability to support evidence-based evaluation. This shows that the manual system is disconnected from institutional assessment needs.

AttendTrack System Evaluation

Table 19: *Functional Suitability*

Statement	Rating 4	Rating 3	Rating 2	Rating 1	WM	SD	Interpretation
Automates attendance recording effectively	22	20	6	2	3.8	0.28	Very Functional
QR code scanning works accurately	21	19	7	3	3.7	0.30	Very Functional
Generates complete attendance reports	20	18	8	4	3.7	0.32	Very Functional
Event evaluation features function properly	18	20	9	3	3.6	0.33	Very Functional
Supports event monitoring	21	19	7	3	3.7	0.29	Very Functional

Legend:

3.26 – 4.00	Very Functional
1.51 – 325	Functional
1.76 – 2.50	Less Functional
1.00 – 1.75	Not Functional

The standout feature of the AttendTrack system was its ability to effectively automate attendance recording, topping the evaluation with a weighted mean of 3.8. Users also expressed high confidence in QR code scanning, report generation, and event monitoring, which all secured a solid weighted mean of 3.7. While event evaluation features scored slightly lower at 3.6, the overall high scores and low variance indicate that the system successfully meets institutional needs for a reliable, streamlined tracking solution.

Table 20: *Performance Efficiency*

Statement	Rating 4	Rating 3	Rating 2	Rating 1	WM	SD	Interpretation
Attendance recording is faster	23	19	6	2	3.8	0.27	Very Efficient
Reports are generated quickly	21	20	7	2	3.7	0.29	Very Efficient
The system reduces workload	20	21	7	2	3.7	0.30	Very Efficient
Real-time monitoring improves efficiency	22	20	6	2	3.8	0.28	Very Efficient
System response time is satisfactory	21	20	7	2	3.7	0.29	Very Efficient

Legend:

3.26 – 4.00	Very Efficient
1.51 – 3.25	Efficient
1.76 – 2.50	Slightly Efficient
1.00 – 1.75	Inefficient

The system excelled particularly in accelerating attendance processes and enhancing operational workflow. The statements Attendance recording is faster and Real-time monitoring improves efficiency both topped the evaluation, securing the highest weighted mean score of 3.8 (SD = 0.27 and SD = 0.28, respectively). This indicates strong consensus among users that the system significantly reduces operational bottlenecks compared to traditional methods.

Table 21: *Usability*

Statement	Rating 4	Rating 3	Rating 2	Rating 2	WM	SD	Interpretation
The system is easy to use	23	19	6	2	3.8	0.27	Very Usable
Navigation is clear	21	20	7	2	3.7	0.29	Very Usable
QR scanning is simple	22	19	7	2	3.8	0.28	Very Usable
Reports are easy to access	21	20	7	2	3.7	0.30	Very Usable
Interface is user-friendly	23	19	6	2	3.8	0.27	Very Usable

Legend:

3.26 – 4.00	Very Usable
1.51– 3.25	Usable
1.76– 2.50	Slightly Usable
1.00 – 1.75	Unusable

The primary benchmarks for user interaction specifically overall ease of use, interface user-friendliness, and the simplicity of the QR scanning mechanism topped the evaluation, each securing a dominant weighted mean score of 3.8 (SD = 0.27 to } 0.28). This high rating indicates that the system's core functionalities are well-designed and computationally accessible. Furthermore, clarity in system navigation and the ease of accessing generated reports both established a solid weighted mean of 3.7 (SD = 0.29 and } 0.30, respectively). The low standard deviations across all metrics confirm a tight consensus among respondents, statistically validating that the platform successfully balances advanced tracking capabilities with a clean, accessible interface suitable for diverse user groups.

***Table 22**: Reliability*

Statement	Rating 4	Rating 3	Rating 2	Rating 1	WM	SD	Interpretation
The system is easy to use	23	19	6	2	3.8	0.27	Very Reliable
Navigation is clear	21	20	7	2	3.7	0.29	Very Reliable
QR scanning is simple	22	19	7	2	3.8	0.28	Very Reliable
Reports are easy to access	21	20	7	2	3.7	0.30	Very Reliable
Interface is user-friendly	23	19	6	2	3.8	0.27	Very Reliable

Legend:

3.26 – 4.00	Very Reliable
1.51 – 3.25	Reliable
1.76 – 2.50	Slightly Reliable
1.00 – 1.75	Unreliable

The system's operational dependability is anchored by its overall ease of use and interface stability, both yielding a peak weighted mean of 3.8 (SD = 0.27). Crucially, the core QR scanning functionality also achieved a weighted mean of 3.8 (SD = 0.28), demonstrating that the system's primary technical mechanism executes tasks predictably and without recurring errors. Furthermore, systemic navigation and report retrieval proved highly dependable, both securing a solid weighted mean of 3.7 The tight clustering of standard deviations (ranging from 0.27 to 0.30) indicates minimal variance among respondent experiences. This statistically confirms that AttendTrack is a stable, robust software solution capable of maintaining long-term data integrity under routine institutional operations.

***Table 23**: Security*

Statement	Rating 4	Rating 3	Rating 2	Rating 1	WM	SD	Interpretation
User login is secure	22	20	6	2	3.8	0.27	Very Secure
Access is limited to authorized users	21	20	7	2	3.7	0.29	Very Secure
Data is protected	22	20	6	2	3.8	0.28	Very Secure
Unauthorized edits are prevented	21	20	7	2	3.7	0.29	Very Secure
Data privacy is maintained	22	20	6	2	3.8	0.27	Very Secure

Legend:

3.26 – 4.00	Very Secure
1.51 – 3.25	Secure
1.76 – 2.50	Less Secure
1.00 – 1.75	Not Secure

The detailed results show that AttendTrack consistently achieved weighted means between 3.6 and 3.8, interpreted as Very Satisfactory across all five dimensions. The raw frequencies confirm that the majority of respondents rated the system at levels 3 and 4, with very few ratings of 1 or 2. Functional suitability, performance efficiency, usability, reliability, and security all received strong approval, with event evaluation features slightly lower but still satisfactory. Overall, AttendTrack was validated as a comprehensive solution that directly addresses the inefficiencies and gaps of the manual process, ensuring event attendance management becomes accurate, efficient, user-friendly, reliable, and secure.

CHAPTER 3

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATION

Summary of findings

This study aimed to design, develop, and evaluate AttendTrack: A Web-Based Event Attendance and Evaluation System for Lapu-Lapu City College as a digital solution to improve event attendance monitoring, reporting, and evaluation. Based on the data gathered from 50 respondents composed of students, event organizers, and administrators, the following findings were obtained.

The findings revealed that the existing manual event attendance system in Lapu-Lapu City College was inefficient, time-consuming, and prone to record management problems. In terms of system inefficiencies, the most common issue identified was that manual attendance recording causes delays (78%), followed by slow verification of participant attendance (50%), while difficulty consolidating attendance data (36%) and increased administrative workload (36%) were also observed. In terms of time consumption, 66% of respondents agreed that attendance tracking was time-consuming, 50% stated that manual monitoring takes too long, and 48% reported delays in updating participant records. Problems in attendance record management were also evident, particularly incomplete attendance records (64%), encoding errors (60%), and lost attendance sheets (52%). Similarly, in monitoring and evaluation, respondents identified lack of real-time attendance monitoring (64%), difficulty tracking participant involvement (56%), and limited access to attendance data (48%) as major concerns. These findings indicate that the current manual system lacks speed, accuracy, and efficiency, making attendance monitoring and event evaluation difficult to manage in an institutional setting.

The results further showed that these problems were rooted in broader system gaps in the current attendance process. In terms of technology and automation gaps, the most common concerns were lack of automation (64%) and manual data consolidation

(64%), while 44% of respondents noted the absence of a QR code system and 28% identified the lack of automated reports. In terms of centralized system issues, 58% reported the absence of a centralized database, while 48% cited the lack of a unified system and limited access to past attendance data. The findings also showed that the manual process was weak in integration and evaluation, as 58% of respondents stated that the current attendance process was not linked to evaluation tools, 54% reported the use of separate systems, 52% identified the absence of feedback collection, and 42% noted limited analytics. These findings confirm that the existing attendance process is not only manual and inefficient, but also technologically outdated, fragmented, and disconnected from the institution's need for real-time monitoring and event assessment.

In response to these identified challenges, AttendTrack was developed and evaluated using the software quality dimensions of ISO/IEC 25010. In terms of functional suitability, the system received very satisfactory to outstanding ratings, with weighted means ranging from 3.6 to 3.8. The highest-rated indicator was "Automates attendance recording effectively" (WM = 3.8), while "QR code scanning works accurately," "Generates complete attendance reports," and "Supports event monitoring" each obtained a weighted mean of 3.7. The event evaluation feature received the lowest weighted mean at 3.6, but it still fell within the strongly agree / very satisfactory range. These findings indicate that AttendTrack successfully delivered the essential functions needed for attendance monitoring, reporting, and event management.

In terms of performance efficiency, the system was also rated very efficient, with weighted means ranging from 3.7 to 3.8. The statements "Attendance recording is faster" and "Real-time monitoring improves efficiency" both received the highest weighted mean of 3.8, while "Reports are generated quickly," "The system reduces workload," and "System response time is satisfactory" each obtained a weighted mean of 3.7. These findings suggest that AttendTrack was effective in improving operational workflow,

reducing delays, and minimizing the workload associated with manual attendance processes.

The system also performed very usable in terms of usability. Respondents rated “The system is easy to use,” “QR scanning is simple,” and “Interface is user-friendly” with weighted means of 3.8, while “Navigation is clear” and “Reports are very accessable” both received a weighted mean of 3.7. These results indicate that the system was perceived as accessible, understandable, and easy to operate even by users with varying levels of experience in digital attendance systems. The low standard deviations across these indicators further suggest a strong consistency in respondents’ positive evaluations of the platform’s usability.

In terms of reliability, AttendTrack likewise rated as very reliable, with the system’s ease of use and interface stability both obtaining a weighted mean of 3.8, while QR scanning also scored 3.8, indicating dependable and stable system operation. Navigation clarity and report accessibility each obtained 3.7, showing that respondents consistently found the system very reliable in carrying out attendance-related tasks without major issues. These findings imply that AttendTrack is capable of maintaining stable performance and preserving attendance records accurately during routine institutional event operations.

Similarly, in terms of security, AttendTrack was rated very secure. “User login is secure,” “Data is protected,” and “Data privacy is maintained” each received a weighted mean of 3.8, while “Access is limited to authorized users” and “Unauthorized edits are prevented” both obtained 3.7. These findings indicate that the system was able to provide secure authentication, controlled access, and data protection mechanisms that are necessary for safeguarding attendance records and event information in the college setting.

Overall, the findings confirmed that AttendTrack was an effective and highly acceptable solution for improving event attendance monitoring and evaluation in Lapu-Lapu City College. Across the five software quality dimensions functional suitability, performance efficiency, usability, reliability, and security the system consistently achieved weighted means ranging from 3.6 to 3.8, interpreted as very satisfactory. The results indicate that the system effectively addressed the major limitations of the manual attendance process by improving attendance recording, reducing administrative workload, increasing data accuracy, enabling real-time monitoring, and supporting more organized event evaluation. These findings validate AttendTrack as a reliable and practical web-based solution for modernizing institutional event management in Lapu-Lapu City College.

Conclusion

Based on the findings of the study, it can be concluded that Lapu-Lapu City College continues to experience several challenges in its existing manual event attendance and evaluation process. These challenges include delays in attendance recording and verification, time-consuming monitoring, incomplete attendance records, encoding errors, lost attendance sheets, and limited access to attendance information during events. The findings also showed that the current process lacks automation, centralized data management, real-time monitoring, and integration between attendance monitoring and event evaluation. As a result, the manual system is no longer sufficient to support efficient, accurate, and organized event management within the institution.

With the design and development of AttendTrack: A Web-Based Event Attendance and Evaluation System**,** these identified challenges were effectively addressed. The system provided a centralized platform for attendance monitoring, QR code-based attendance recording, automated report generation, and event evaluation. Through these features, AttendTrack improved the speed and accuracy of attendance tracking, reduced the administrative burden associated with manual monitoring, and allowed users to access

attendance information more efficiently. The system also supported better event management by enabling real-time monitoring of participants and more organized handling of event records.

In terms of software quality, the results showed that AttendTrack performed very satisfactorily across the five ISO/IEC 25010 quality dimensions used in the study, namely functional suitability, performance efficiency, usability, reliability, and security. The system successfully delivered the required functions for attendance recording, report generation, and event evaluation; improved the efficiency of attendance-related tasks; provided a user-friendly and accessible interface; maintained stable and dependable performance during use; and ensured secure access and protection of attendance records. These qualities indicate that the system is suitable for actual institutional use and can serve as a practical tool for modernizing event attendance management in Lapu-Lapu City College.

Moreover, the respondents expressed a highly positive perception of the system, as reflected in the very satisfactory weighted mean results across all evaluated criteria. The strongest aspects of the system included its ability to automate attendance recording, improve real-time monitoring, simplify QR code scanning, and maintain data security and accessibility. Although there are still opportunities for future enhancement, such as expanding system accessibility and adding more advanced features, the overall findings confirm that AttendTrack is an effective and reliable solution for improving event attendance and evaluation processes. Therefore, the study concludes that the system successfully achieved its objectives and addressed the research problems by identifying the weaknesses of the manual process, providing a digital solution to these issues, and evaluating its performance based on recognized software quality standards.

Recommendations

Based on the findings and conclusion, several recommendations are proposed to further strengthen the implementation and impact of AttendTrack:

Lapu-Lapu City College should adopt AttendTrack across all departments to standardize event attendance monitoring and evaluation, ensuring consistency and efficiency throughout the institution.

Future development should consider enhancements such as mobile application integration, offline attendance capability, and biometric authentication to further improve usability and security.

Large-scale performance testing should be conducted to ensure the system's reliability under high concurrent usage, particularly during major institutional events.

Future evaluations should include administrators and event organizers to capture diverse perspectives and feedback, thereby enriching the system's adaptability to different user roles.

Continuous improvement through iterative updates based on user feedback is essential to maintain system relevance and effectiveness.

OUTPUT OF THE STUDY

ATTENDTRACK: A WEB-BASE EVENT ATTENDANCE AND EVALUATION SYSTEM IN LAPU-LAPU CITY COLLEGE

RATIONALE

This study introduced AttendTrack, a fully integrated web-based event attendance and evaluation system developed to address the identified gaps in Lapu-Lapu City College's existing manual attendance processes. The findings revealed that traditional attendance methods are inefficient, time-consuming, and susceptible to human error, leading to inaccurate records and delays in report generation. Through the implementation of QR code-based attendance tracking and real-time monitoring, the system improved process efficiency, reduced manual intervention, and enhanced the accuracy and reliability of attendance data.

The system improved usability and accessibility by providing a centralized platform that integrates event management, participant registration, attendance monitoring, and evaluation functionalities. Its user-friendly interface, role-based access control, and real-time dashboards enabled administrators, organizers, and students to effectively manage and monitor event participation. Furthermore, the inclusion of automated reporting and evaluation features facilitated faster data processing and supported the collection of participant feedback, strengthening event assessment and institutional decision-making.

To ensure system security and reliability, AttendTrack incorporated secure authentication mechanisms, controlled user access, and protected data management processes. The results of the study confirmed that the system effectively addressed the limitations identified in the Statement of the Problem (SOP), particularly in improving operational efficiency, data accuracy, real-time monitoring, and evaluation processes. Overall, AttendTrack delivers a cohesive, functional, and scalable solution tailored to real-world academic needs, supporting digital transformation in institutional event management.

OBJECTIVES OF THE SYSTEM

General Objective

The primary objective of the AttendTrack system is to develop a web-based event attendance and evaluation platform that streamlines event management through QR code technology, real-time attendance monitoring, automated reporting, and event evaluation. The system is designed to provide an accurate, secure, and user-friendly solution that improves institutional event management while complying with the ISO/IEC 25010 software quality standards.

Specific Objectives

1. To develop a QR code-based attendance feature that enables fast, accurate, and paperless attendance recording during institutional events.
2. To provide a centralized event management module that allows administrators and organizers to create, update, and manage event information efficiently.
3. To implement real-time attendance monitoring that enables organizers to track participant attendance instantly through an interactive dashboard.
4. To establish a secure role-based access control system that assigns appropriate permissions to students, event organizers, and administrators.
5. To integrate an event evaluation module that allows participants to submit feedback, helping organizers assess the effectiveness of each event.
6. To generate automated attendance summaries and evaluation reports that support documentation, monitoring, and informed decision-making.
7. To evaluate the system based on the ISO/IEC 25010 software quality model, focusing on functional suitability, performance efficiency, usability, reliability, and security.
8. To develop a scalable and maintainable web-based system that can support future enhancements and the growing event management needs of Lapu-Lapu City College.

TECHNICAL REQUIREMENTS OF ATTENDTRACK SYSTEM

Table 24: *Hardware and Software Requirements*

CATEGORY	HARDWARE			SOFTWARE	
Client-Side Requirements	**Component**	**Minimum Specs**	**Recommended Specs**	**Requirements**	**Description**
	Processor	Dual-core 1.8 GHz	Quad-Core 2.5 GHz or Higher	Web browser	Chrome, Firefox , Edge
	RAM	4 GB	8 GB or More	Operating System	Windows 10/11 macOS Android iOS
	Storage	500 MB	10 GB or More	Frontend	HTM, CSS, JS
	Display	1366x768	1920x1080	Internet	Required for real-time Access
Server-Side Requirements	Network	Stable 1mbps**+**	Stable 10mbps+	Browser Extensions	Exportable Reports, If Generated in PDF Format and QR Scanning
	Device	Smart Phone/PC	Smartphone with Camera		
	Processor	Dual-Core 2.0 GHz	Quad-Core 3.0 GHz +	Operating System	Windows Server/Linux
	RAM	4GB	8 GB +	Web Server	Apache or Nginx
	Storage	40GB SSD	100GB SSD+	Programming Language	
	Network	10 Mbps	20 Mbps+	Database	MySQL/PostgreSQL

The AttendTrack: A Web-Based Event Attendance and Evaluation System requires specific hardware and software to ensure efficient operation. The client-side requirements include compatible web browsers, sufficient memory, processing capability, and stable internet connection for real-time attendance and QR code scanning. Meanwhile, the server-side requirements involve higher processing power, storage capacity, and a reliable operating system to support system processing, data management, and report generation. These requirements ensure the system's reliability and accessibility across different devices.

System Architecture

The AttendTrack system follows a three-tier architecture consisting of the presentation, application, and data layers. The presentation layer allows users to access the system through web browsers, while the application layer handles core functions such as authentication, attendance tracking, and report generation. The data layer manages and stores all system information, including user, event, and attendance records. This structure ensures efficient, secure, and reliable system performance.

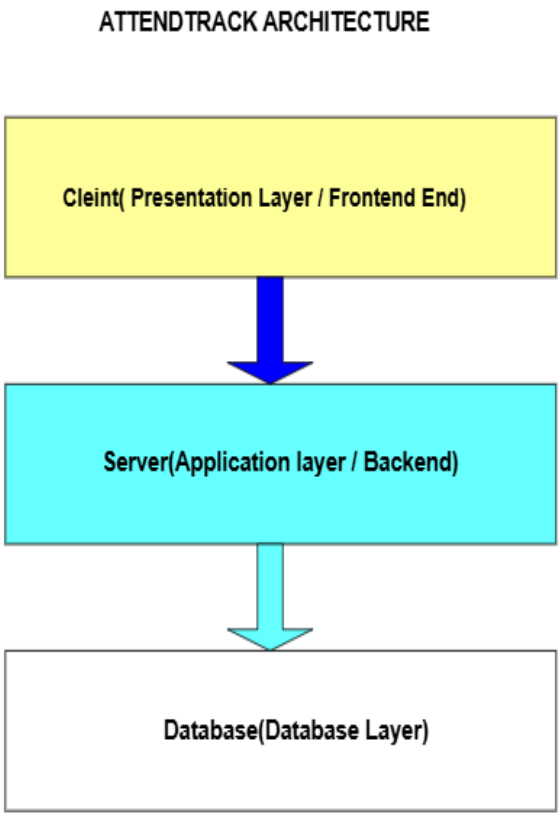


Figure 4: Modified Three-Tier Architecture

As shown in Figure 4 Attendtrack System Modified Three layer architecture, the system is organized into Three main layers:

Presentation Layer (Client-Side)

This is the part of AttendTrack that users actually see and use. It's the interface where students check in, organizers manage events, and admins monitor attendance. Researchers made it user-friendly and simpleyou don't need to install anything, just open it on a modern browser like Chrome or Edge. As long as the device has a decent screen (at least 1366×768), it works perfectly.

Application Layer (Server-Side / Backend)

Think of this layer as the brain of AttendTrack. Everything that happens like recording attendance or sending notifications goes through here. Researchers built it using Laravel and PHP, following a structured approach called MVC to keep everything organized. This layer handles:

Data Layer (Database)

This is where all the information is safely stored. Every event, participant registration, and attendance record are kept here. Researchers use MySQL or MariaDB for reliable data storage.

System Workflow

The AttendTrack: A Web-Based Event Attendance and Evaluation System in Lapu-Lapu City College, follows a structured workflow that facilitates real-time attendance tracking, secure data processing, and automated reporting. The system is built on a web-based architecture using a modern tech stack (e.g., PHP, JavaScript, and MySQL) and securely handles client-server interactions over HTTPS. The workflow consists of four key steps, as outlined below:

1. User Request via HTTPS

The workflow primarily begins when users such as (Student, Admin and Organizer) interact with the system via a web browser. These interactions involve sending requests to the server, including but not limited to:

Authenticating into the administrative or student portal

Initializing events and configuring attendee master lists.

Executing real-time scans (QR code or ID input) to capture attendance.

Generating and exporting institutional attendance reports.

2. Server-Side Processing and Validation

Once a request is received, the backend validates the input to maintain data accuracy. The system then authenticates the scanned ID by querying the Lapu-Lapu City College student records. This phase includes:

Verification of Identity: To ensure that the student is currently enrolled and can enter the portal the event,

Duplicate Prevention: Checking if a "Time-In" status already exists for that specific user to prevent double-logging.

Logic Execution: Calculating "On-Time" or "Late" status based on the server's atomic clock.

3. Database Persistence

Once it will validate, the transaction will go to the database. The system updates the relational tables to reflect the attendee's new status. This ensures Data Persistence, meaning the attendance record is permanently stored and protected against accidental data loss or unauthorized manual tampering after the event window closes.

Product Backlog

This backlog provides a structured approach in developing AttendTrack: A Web-Based Event Attendance and Evaluation System in Lapu-Lapu City College**,** ensuring efficient event handling, attendance tracking, and user management.

Table 25: Product Backlog

Feature	User Story	Task	Start Date	End Date	Week Duration
Admin Dashboard	As an admin, I want to view overall system activity to monitor events and attendance.	View total events created; Monitor attendance summaries; View active users	23-Aug-25	25-Aug-28	1 Week
Event Management	As an admin, I want to create and manage events to organize activities efficiently.	Create/edit/delete events; Set event details (date, time, venue); Assign event coordinators	30-Aug-25	05-Sep-25	1 Week
Attendance Monitoring	As an admin, I want to track attendance records to ensure accurate participation data.	View attendance logs; Filter attendance by event; Generate attendance summaries	06-Sep-25	12-Sep-25	1 Week
User Management	As an admin, I want to manage users to control system access.	Add/edit/delete users; Assign roles (admin, staff, participant); Activate/deactivate accounts	13-Sep-25	19-Sep-25	1 Week
QR Code System	As a staff member, I want to scan participants to record attendance quickly.	Generate QR codes Scan QR codes; Validate attendance entries	20-Sep-25	26-Sep-25	1 Week
Real-Time Attendance Tracking	As a staff member, I want to see attendance in real-time to monitor participation.	Display live attendance count; Show present/absent participants; Update dashboard instantly	27-Sep-25	03-Oct-25	1 Week
Event Registration	As a participant, I want to register for events easily.	Online registration form; Store participant details; Confirm registration	04-Oct-25	10-Oct-25	1 Week
Participant Management	As an admin/staff, I want to manage participant data for each event.	View participant list; Edit participant info; Remove or transfer participants	11-Oct-25	17-Oct-25	1 Week
Reports Generation	As an admin, I want to generate reports to analyze attendance and event performance.	Generate attendance reports; Export reports (PDF/CSV); View event analytics	18-Oct-25	24-Oct-25	1 Week
Notification System	As a user, I want to receive notifications about events and updates.	Send event reminders; Notify registration confirmation; Send announcements	25-Oct-25	31-Oct-25	1 Week

Feature	User Story	Task	Start Date	End Date	Week Duration
Role-Based Access Control	As a system admin, I want to restrict access based on roles for security.	Define permissions; Restrict features per role; Manage role access levels	01-Nov-25	07-Nov-25	1 Week
Event History Archive	As an admin, I want to store past events for reference and reporting.	Archive completed events; View historical data; Retrieve past attendance records	08-Nov-25	14-Nov-25	1 Week
System Security	As an admin, I want to ensure data security and integrity.	Implement login authentication; Encrypt sensitive data; Maintain audit logs	15-Nov-25	21-Nov-25	1 Week
Data Backup & Recovery	As a system admin, I want to protect system data from loss. As a system admin, I want to protect system data from loss.	Schedule automatic backups; Restore data when needed; Maintain backup logs	22-Nov-25	28-Nov-25	1 Week
User Interface & Experience	As a user, I want a simple and intuitive interface for ease of use.	Design responsive UI; Improve navigation; Optimize user workflows	28-Mar-26	03-Apr-26	1 Week
System Performance Optimization	As a user, I want the system to run smoothly and quickly.	Optimize database queries; Reduce loading time; Improve system responsiveness	05-Apr-26	11-Apr-26	1 Week

SYSTEM PROGRAM FLOW

The AttendTrack system operates through a structured workflow that integrates both administrative control and student interaction to ensure efficient event and attendance management. The process begins with secure user authentication, directing admins and students to their respective dashboards. Administrators handle the core setup by creating and managing events, organizing student and course data, and preparing the system for attendance tracking. During events, QR code technology is utilized, allowing students to scan their unique codes for quick and accurate attendance recording. The system then automatically updates attendance records in real time, which can be reviewed, managed, and analyzed by the admin through report generation and system monitoring tools. Meanwhile, students can view attendance updates, join events, and receive notifications, ensuring active participation and awareness. Overall, the workflow streamlines operations, minimizes manual effort, and enhances accuracy, transparency, and efficiency in managing attendance and events

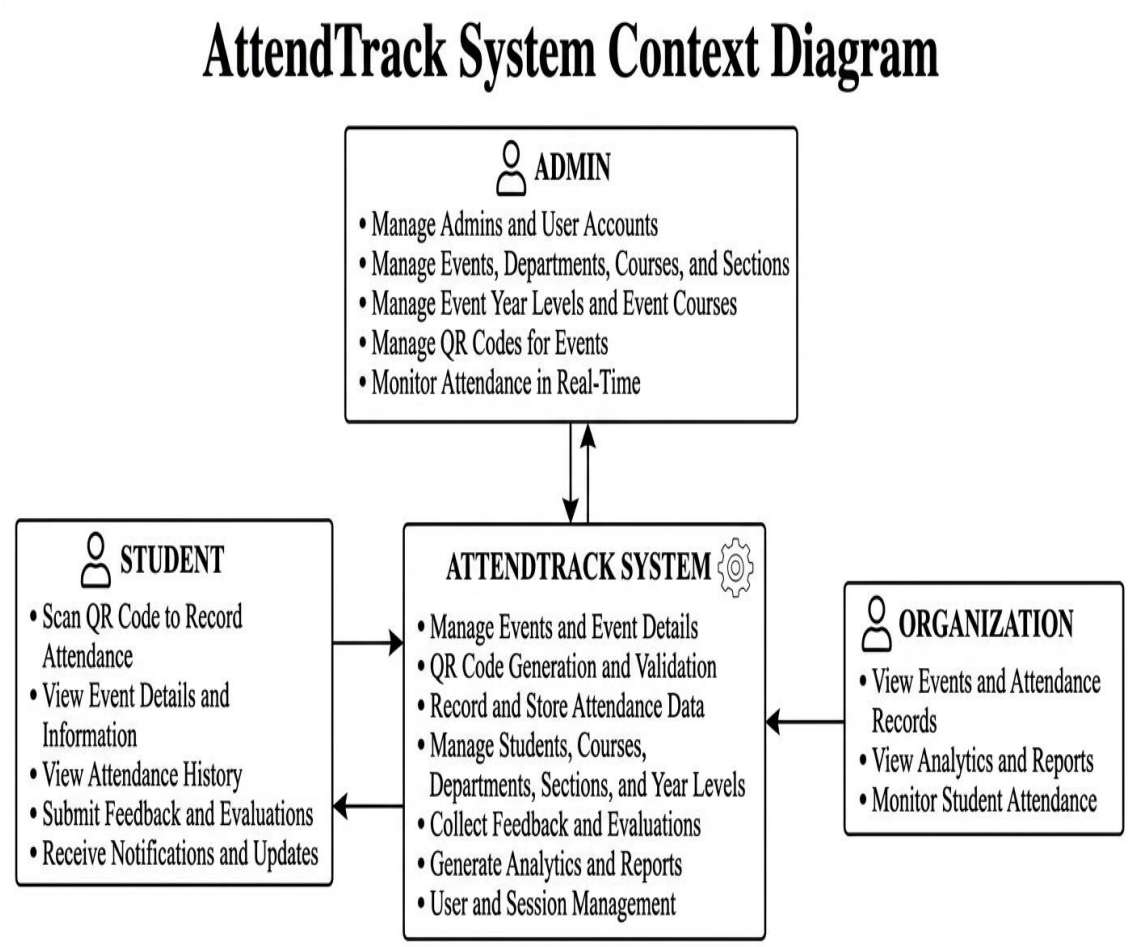


Figure 5: *Context Diagram*

Context Diagram

The Context Diagram shows how the AttendTrack System interacts with its main users: the Admin, Student, and Organization. The Admin manages user accounts, creates and organizes events, assigns courses and departments, generates QR codes, and monitors attendance. The Student scans QR codes for attendance, views event information and attendance records, submits evaluations, and receives notifications. The Organization monitors student participation, checks attendance records, and reviews reports and analytics. Overall, the diagram shows how AttendTrack connects these users in one centralized system to make event management, attendance monitoring, and evaluation easier and more organized.

System Workflow

This diagram depicts the step-by-step workflow of the users

Super Admin User Flow

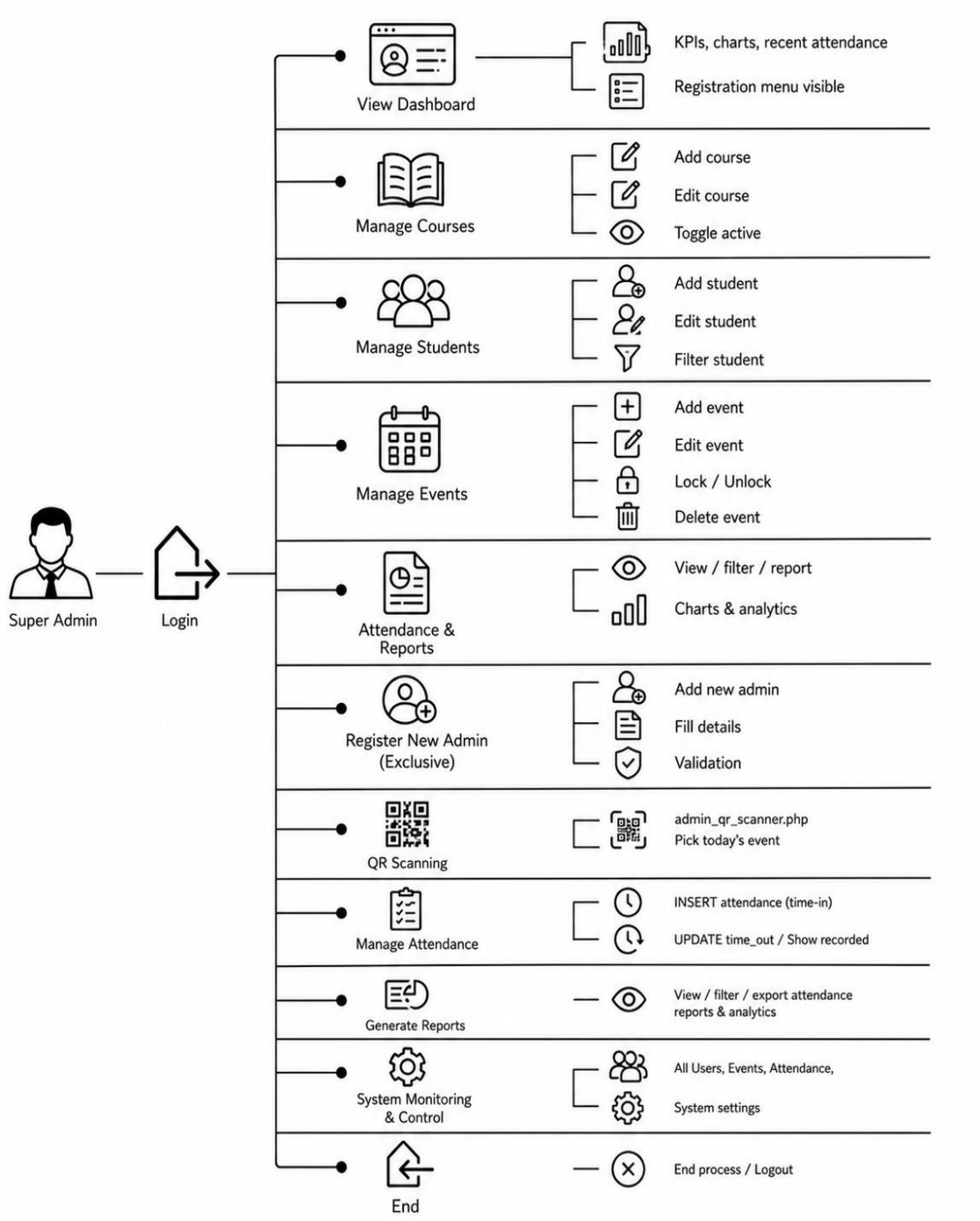


Figure 6: *Super Admin User Flow*

Super Admin User Flow

In figure 8 illustrates of the Super Admin workflow in the AttendTrack System depicts a structured process that starts with secure login and verification, granting access to the dashboard where key indicators, recent attendance data, and system controls are displayed. From this main interface, the Super Admin can efficiently handle core functions such as managing courses, students, and events, including adding, updating, locking, or removing records. The workflow also highlights access to attendance and reporting features for tracking and analyzing data. A notable component shown in the diagram is the registration of new admin accounts through a validated input procedure. Additionally, the system incorporates QR code scanning for real-time attendance recording, allowing entries for time-in, time-out, and detection of duplicate scans. The process continues with attendance management, report generation, and overall system monitoring, enabling full supervision of users, events, and records, before ending the workflow.

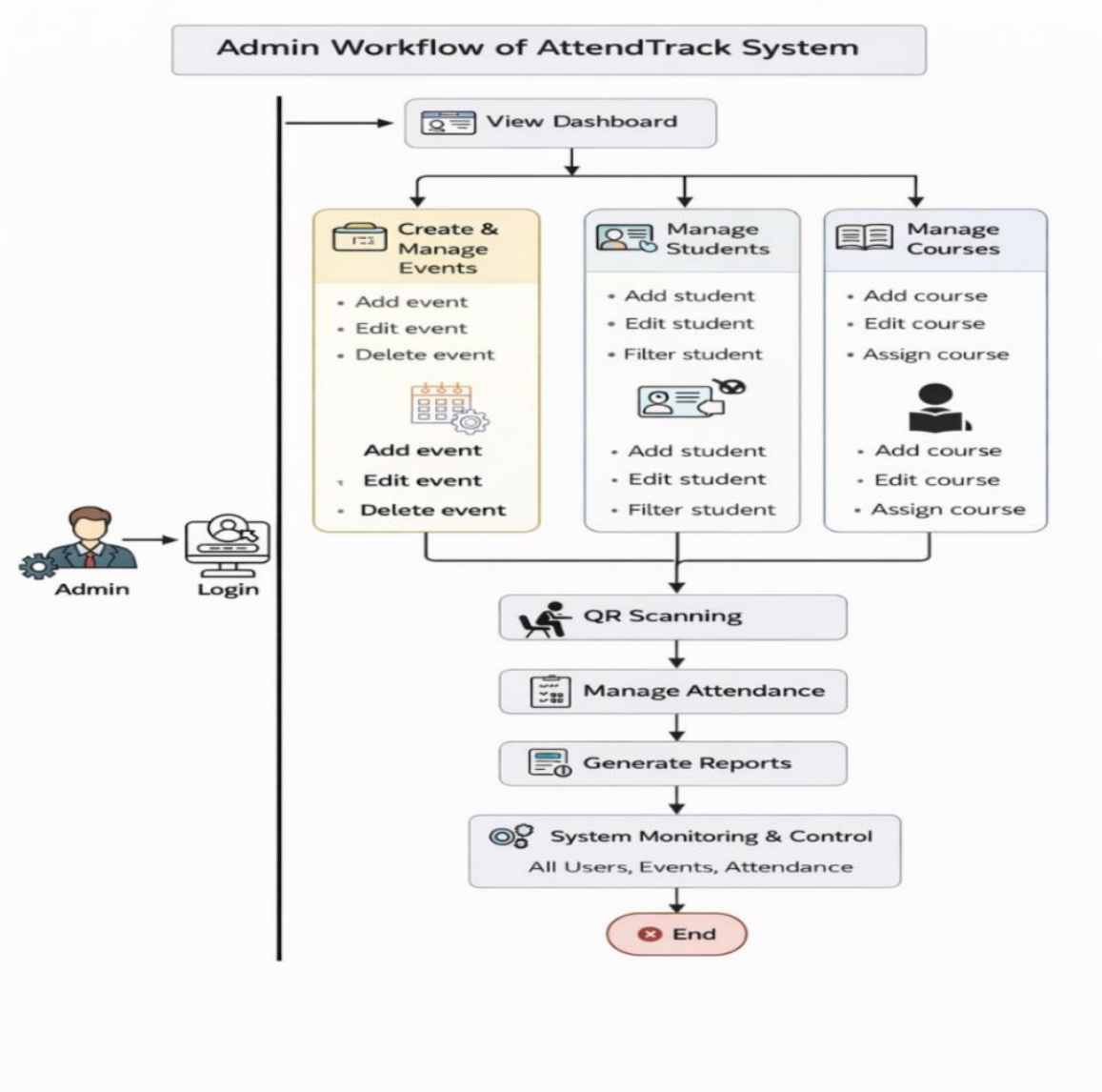


Figure 7: Admin User Flow

Admin User Flow

The figure illustrates the Admin User Flow of the AttendTrack system, a centralized platform where the administrator begins each session with secure login to access a comprehensive dashboard. From this dashboard, the admin can efficiently manage core system components, including event management (creating, editing, and deleting events), student management (adding, updating, and filtering student records), and course

management (organizing and assigning courses). The workflow continues with QR code scanning for accurate attendance tracking, followed by attendance management to review and validate records. The admin can also generate detailed reports for monitoring and evaluation purposes, while system monitoring and control ensure full visibility over users, events, and attendance data. This structured workflow enhances administrative efficiency, reduces manual processes, and ensures accurate and organized event and attendance management.

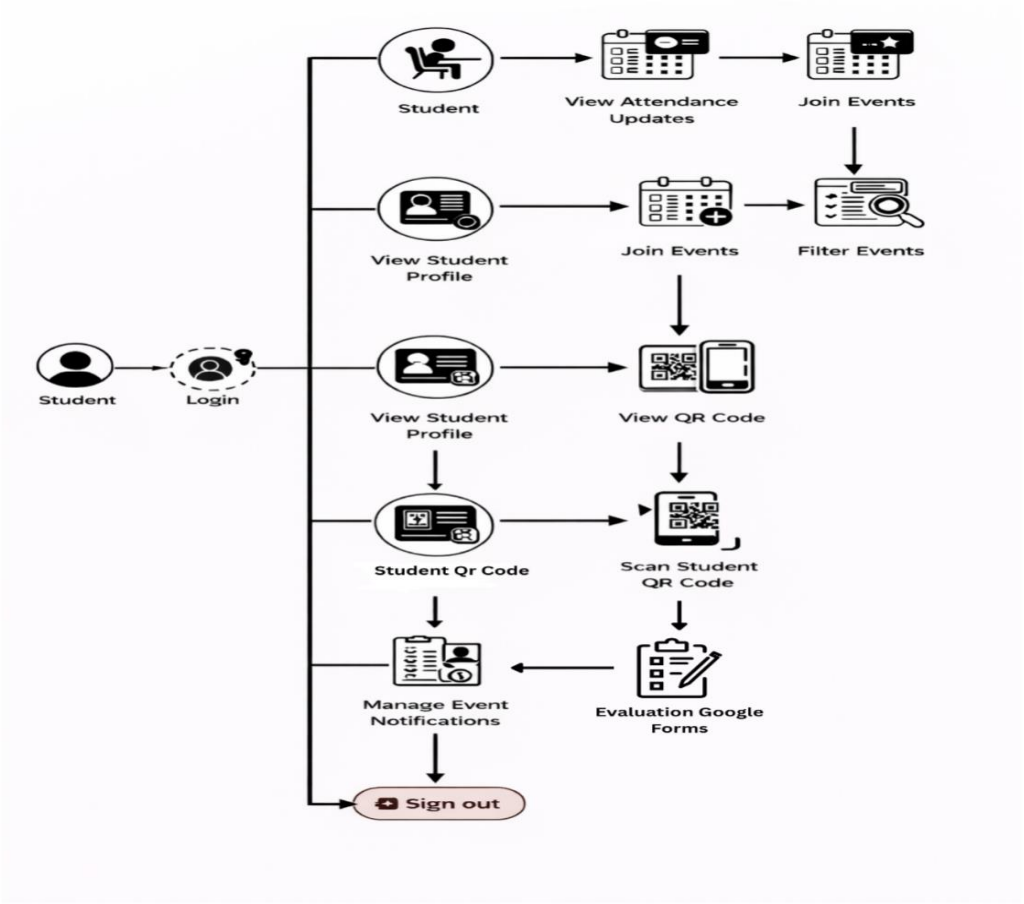


***Figure 8**: Student User Flow*

Student User Flow

The student user flow starts with login, allowing access to personal features within the system. After logging in, the student can view and update their profile, check attendance updates, and explore available events. Students can join events and use filtering options to easily find relevant activities. They can also generate or view their QR code, which is used for attendance scanning during events. The system provides notifications related to events and attendance updates to keep students informed. Finally, after completing their tasks, the student can sign out of the system, completing the workflow.

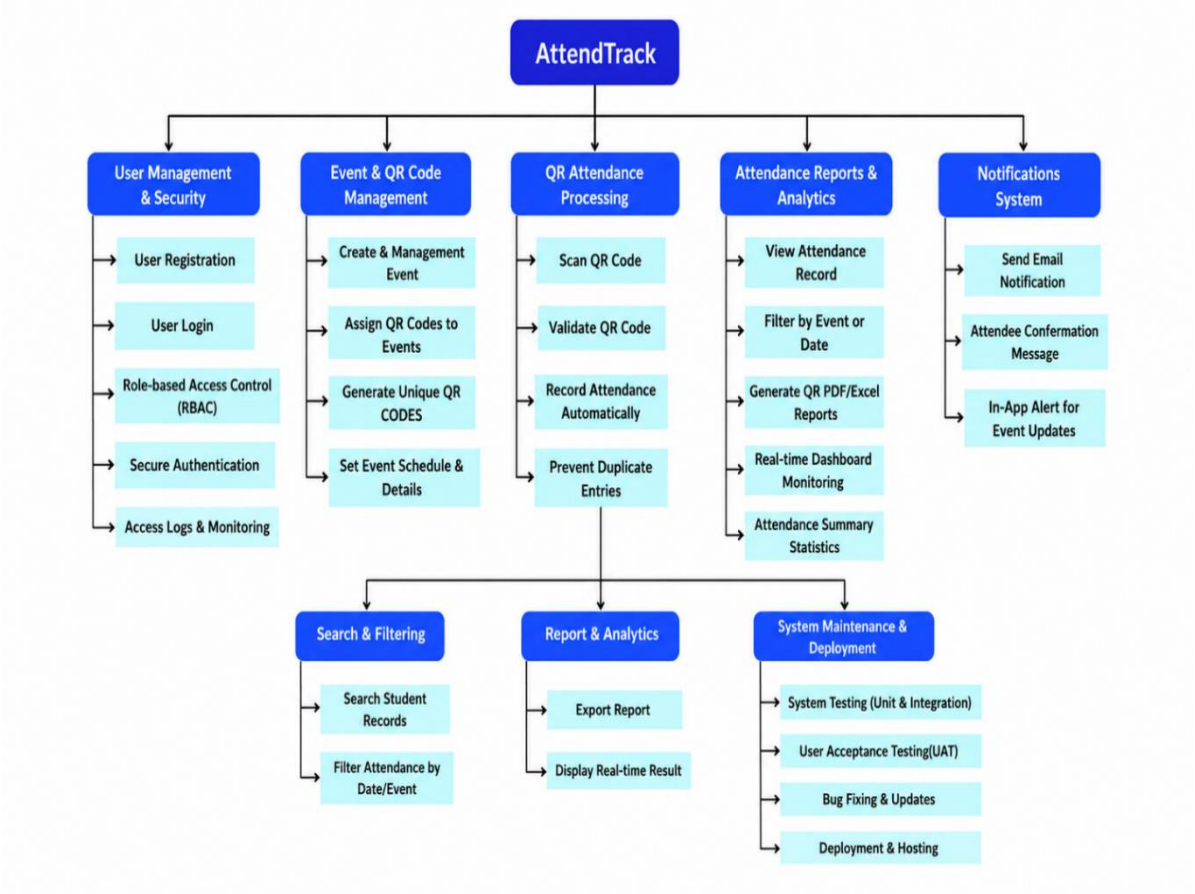


Figure 9*: Functional Decomposition Diagram*

Functional Decomposition Diagram

The Functional Decomposition Diagram provides a clear overview of how the AttendTrack system is organized by breaking the system down into its main functional modules and the specific tasks performed within each module. Instead of viewing the system as a single application, the diagram shows how its features work together to support event attendance and evaluation. The major modules include User Management and Security, Event and QR Code Management, QR Attendance Processing, Attendance Reports and Analytics, Notifications System, Search and Filtering, Report and Analytics, and System Maintenance and Deployment. Each module contains related functions, such as user authentication, event creation, QR code generation, attendance recording, report generation, notification delivery, and system maintenance. By presenting the system in this structured manner, the diagram makes it easier to understand how each component contributes to the overall operation of AttendTrack. It also highlights the logical organization of the system, ensuring that all features work together to provide an efficient, secure, and reliable platform for managing event attendance and evaluation.

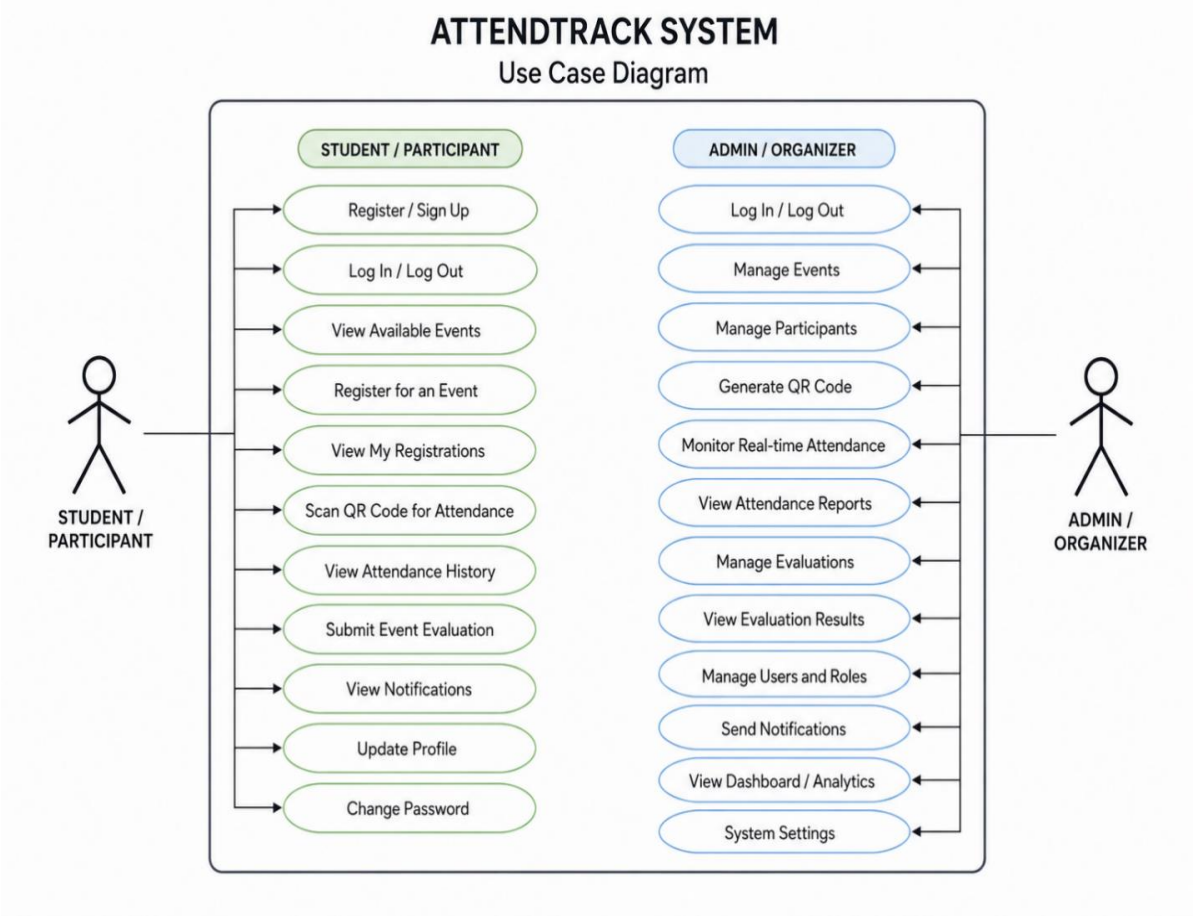


Figure 10: Use Case Diagram

Use Case Diagram

The Use Case Diagram presents the functional interactions between the users and the AttendTrack System at Lapu-lapu City College, It defines the system scope and role-based access. The diagram features two main actors with distinct duties. The Students/Participant registers accounts, logs in, and join events. They also scan Qr codes, check records and submit evaluations. The Admin/Organizer manages overall platform operations. They handle event records,generate Qr codes, and monitoring live attendance. These clear user roles ensure secure, efficient event management and tracking

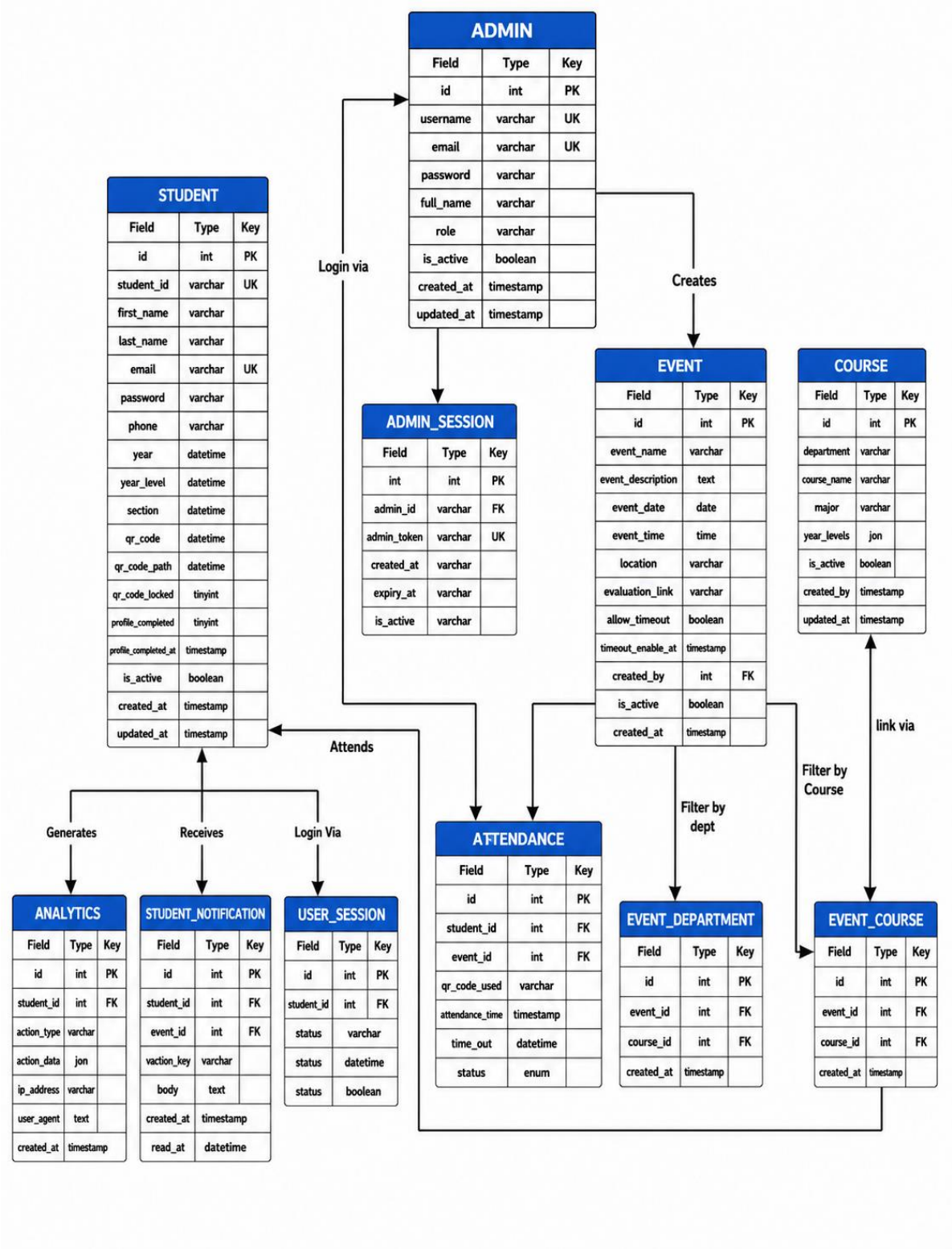


Figure 11: *Entity-Relationship Diagram (ERD)*

Entity–Relationship Diagram

The Entity-Relationship Diagram (ERD) of the AttendTrack: A Web-Based Event Attendance and Evaluation System present the overall structure of the database and shows how the different entities are connected. It illustrates how data is organized and how each part of the system works together to support event management, attendance monitoring, user authentication, and communication between administrators and students.

The **Admin** entity stores the information of system administrators, including their username, email address, password, role, account status, and record timestamps. Administrators are responsible for managing the system by creating events, organizing courses, and maintaining other important records. To improve security, each administrator is connected to the **Admin Session** entity, which keeps track of active login sessions, authentication tokens, and session expiration.

The **Student** entity contains the personal and academic information of registered students. This includes their student ID, name, email address, contact number, course, year level, section, QR code information, and account status. These records allow the system to uniquely identify every student and manage their participation in school events.

The **User Session** entity records each student's login session whenever they access the system. By storing session information, the system can verify user authentication and ensure that only authorized students can access its features.

The **Event** entity serves as the main record for all events created within the system. It stores essential information such as the event name, description, schedule, venue, evaluation link, attendance timeout settings, and the administrator who created the event. This allows administrators to efficiently organize and manage different school activities.

The **Attendance** entity records every student's attendance during an event. It links a student to a specific event while storing details such as the QR code used, attendance time, timeout record, and attendance status. This enables the system to maintain accurate attendance records and generate reliable reports.

The **Course** entity contains the academic programs offered by the institution, including the department, course name, major, year levels, and status. This information helps categorize students according to their academic program and allows administrators to organize events for specific groups.

The relationship between events and courses is managed through the **Event Course** entity. This allows an event to be assigned to one or more courses, ensuring that only the intended students can participate. Likewise, the **Event Department** entity links events to specific departments, making it easier to organize institution-wide or department-exclusive activities.

The **Student Notification** entity stores notifications sent to students regarding upcoming events, reminders, announcements, and other important updates. It also records whether a notification has been read, helping ensure that students receive timely information about school activities.

Lastly, the **Analytics** entity records user activities performed within the system, such as logins, attendance submissions, and other important actions. These records help administrators monitor system usage, review user activity, and support future system improvements.

Overall, the ERD provides a clear and organized database structure for the AttendTrack system by organizing administrator, student, event and records. This design ensures efficient data management to support reliable event attendance monitoring and evaluation

TEST RESULTS

Table 26: *Test Results*

Test ID	Test Scenario	Test Steps	Pre-Requisites	Test Data	Test Status
ATTC01	Verify administrator login	1. Open login page. 2. Enter valid admin credentials. 3. Click **Login**.	Admin account is registered.	Username: superadmin@llcc.edu.ph Password: SuperAdmin@123	PASS
ATTC02	Verify student login	1. Open login page. 2. Enter valid student credentials. 3. Click **Login**.	Student account exists.	Student ID: 2026-0001	PASS
ATTC03	Verify invalid login	1. Enter incorrect username/password. 2. Click **Login**.	User account exists.	Invalid credentials	PASS
ATTC04	Verify event creation	1. Login as Admin. 2. Select **Create Event**. 3. Save event.	Administrator logged in.	Event: IT Week 2026	PASS
ATTC05	Verify event editing	1. Open event details. 2. Modify information 3. Save changes.	Existing event available.	Updated venue and schedule	PASS
ATTC06	Verify event deletion	1. Select event. 2. Click **Delete**. 3. Confirm deletion.	Existing event available.	Event ID: EVT-001	PASS
ATTC07	Verify QR code generation	1. Open event. 2. Generate QR Code.	Event successfully created.	Event QR Code	PASS
ATTC08	Verify QR attendance scanning	1. Student scans QR code. 2. System validates attendance.	Active event and valid QR code.	Student QR Code	PASS

Test ID	Test Scenario	Test Steps	Pre-Requisites	Test Data	Test Status
ATTC09	Verify duplicate attendance prevention	1. Scan the same QR code twice.	Attendance already recorded.	Duplicate QR scan	PASS
ATTC10	Verify attendance record display	1. Open Attendance module. 2. View participant list.	Attendance data available.	Event Attendance Log	PASS
ATTC11	Verify event evaluation submission	1. Open evaluation link. 2. Answer questionnaire. 3. Submit responses.	Student attended the event.	Evaluation responses	PASS
ATTC12	Verify notification delivery	1. Create event. 2. Send notification.	Student accounts available.	Event Reminder	PASS
ATTC13	Verify report generation	1. Open Reports. 2. Generate attendance report.	Attendance records exist.	Attendance Summary	PASS
ATTC14	Verify analytics dashboard	1. Login as Admin. 2. Open Dashboard.	Attendance data available.	Dashboard statistics	PASS
ATTC15	Verify course assignment to event	1. Create event. 2. Assign course.3. Save.	Course records available.	BSIT 4th Year	PASS
ATTC16	Verify department filtering	1. Select department filter. 2. View available events.	Department assigned to event.	College of Technology	PASS
ATTC17	Verify student profile update	1. Open Profile. 2. Edit information. 3. Save changes.	Student logged in.	Updated contact number	PASS
ATTC18	Verify logout function	1. Click **Logout**. 2. Return to login page.	User is logged in.	N/A	PASS

Software Requirements

The development of AttendTrack relies on carefully chosen hardware and software to ensure optimal functionality and security.

Front End Development

HTML5 + CSS3 (custom stylesheets in assets/css/).

JavaScript (vanilla).

Font Awesome 6.0 / 6.4 — icons (CDN).

Google Fonts — Roboto, Inter, Poppins (CDN).

html5-qrcode — in-browser QR scanning (camera) for admin_qr_scanner.php.

FullCalendar 6.1.10 — event calendar UI in admin_create_event.php.

Chart.js 4.4.1 — analytics/KPI charts on the admin dashboard.

jsPDF 2.5.1 + jspdf-autotable 3.5.31 — PDF export of attendance reports.

Back End Development

PHP 7.4+ / 8.x (server-side language).

MySQL / MariaDB (relational database — event_attendance).

Apache HTTP Server (web server).

PDO MySQL extension (database access in config/database.php).

GD library / extension (used by includes/qr_generator.php for embedding the LLCC logo into QR images: image create from png, image create from jpeg, etc.). api.qrserver.com — external QR-image generation API.

Authentication Tools

PHP password_hash() with PASSWORD_DEFAULT (bcrypt) for storing passwords. password_verify() for login validation.

Session cookies (admin_session, user_session) backed by the admin_sessions and user_sessions tables (24-hour expiry) .

random_bytes() + bin2hex() for cryptographically secure session tokens.

hash_hmac('sha256', …) for the hard-coded super-admin session token.

Role-based access control (admin vs super_admin) enforced in includes/admin_auth.php and includes/auth.php.

Development Tools

XAMPP — bundles Apache + MySQL + PHP (project lives in htdocs/EventAttendance).

phpMyAdmin — database administration / SQL execution (e.g. running database.sql, create_admin_tables.sql, etc.).

Code editor — VS Code / Cursor (project has a .vscode/ folder).

Web browser — Chrome / Firefox / Edge for testing the UI and QR scanner (camera permission required).

Git — version control (optional, project structure suggests source control use).

Web server:

Z.com for hosting application.

Security Features:

SSL/TLS Certificates for encrypting data transfers.

Regular software patching for security and performance enhancements.

Operating Systems Supported

Server side: Windows, Linux, or macOS — anywhere XAMPP (or LAMP/MAMP/WAMP) can run.

Client side (browser-based, fully responsive.

Desktop: Windows, macOS, Linux.

Mobile: Android (Chrome) and iOS (Safari) — used by students for QR display and by admins for QR scanning via the device camera.

Hardware Requirements

1. **Server-Side Requirements (Local Server / XAMPP Deployment)**

Compute Environment:

Server Type: Local host machine running XAMPP (Apache, MySQL, PHP)

Operating System: Windows 10/11, macOS, or Linux (Ubuntu recommended).

Processor:

Minimum: Dual-core 2.0 GHz (Intel Core i3 / AMD Ryzen 3).

Recommended: Quad-core 2.5 GHz+ (Intel Core i5 / AMD Ryzen 5 or higher).

Memory (RAM):

Minimum: 4 GB.

Recommended: 8 GB or more.

Storage:

Minimum: 5 GB free space.

Recommended: 20 GB SSD (for faster database queries and QR processing).

Security:

Protocol Requirement: HTTPS or localhost required for camera-based QR scanning.

Access Control:Role-based authentication for admin and users.

Data Protection: Secure session handling and login authentication mechanisms.

Networking:

Connection: Stable internet connection for CDN libraries (e.g., Chart.js, Font Awesome, jsPDF, html5-qrcode, FullCalendar).

Deployment: Local Area Network (LAN) or hosted online server.

IP Configuration: Static local IP recommended for multi-device access.

Database:

Type: MySQL (via XAMPP).

Storage Allocation: Depends on attendance records, QR codes, and uploaded files.

System Constraints:

All devices must be connected to the same network (LAN) if locally hosted.

Browser camera access requires secure origin (HTTPS or localhost).

Fully web-based system (no installation required on client devices).

2. **Client-Side Requirements**

Desktop/Laptop (Administrator / Organizer Device).

Processor: Dual-core 2.0 GHz or higher.

RAM: Minimum 2 GB (4 GB recommended).

Display: Minimum 720p (1080p recommended for dashboard use).

Camera: Required (for QR code scanning functionality).

Browser: Latest versions of Chrome, Edge, Firefox, or Safari.

Network Interface: Wi-Fi or Ethernet connection to the server.

Mobile Devices (Administrator / Scanner Use)

Operating Systems: Android or iOS (latest versions recommended).

Camera: Required (rear camera preferred for QR scanning).

RAM: Minimum 2 GB.

Browser: Modern mobile browser (Chrome, Safari, etc.).

Connectivity: Wi-Fi or mobile data connected to the same network/server.

Student Devices

Device Type: Smartphone, tablet, laptop, or desktop.

Operating Systems: Android, iOS, Windows, macOS.

Browser: Any modern web browser.

Screen Resolution: Minimum 360 px width (responsive UI supported).

Camera: Not required (students only display QR codes).

RAM: Minimum 1 GB.

Connectivity: Wi-Fi or mobile data.

Networking and Peripheral Requirements

Wi-Fi Router / Access Point: Required for connectivity within the event venue.

Printer (Optional): For printing QR codes if needed.

UPS / Power Backup (Optional): Prevents data loss during system operation.

Projector/External Display (Optional):For live dashboard monitoring during events.

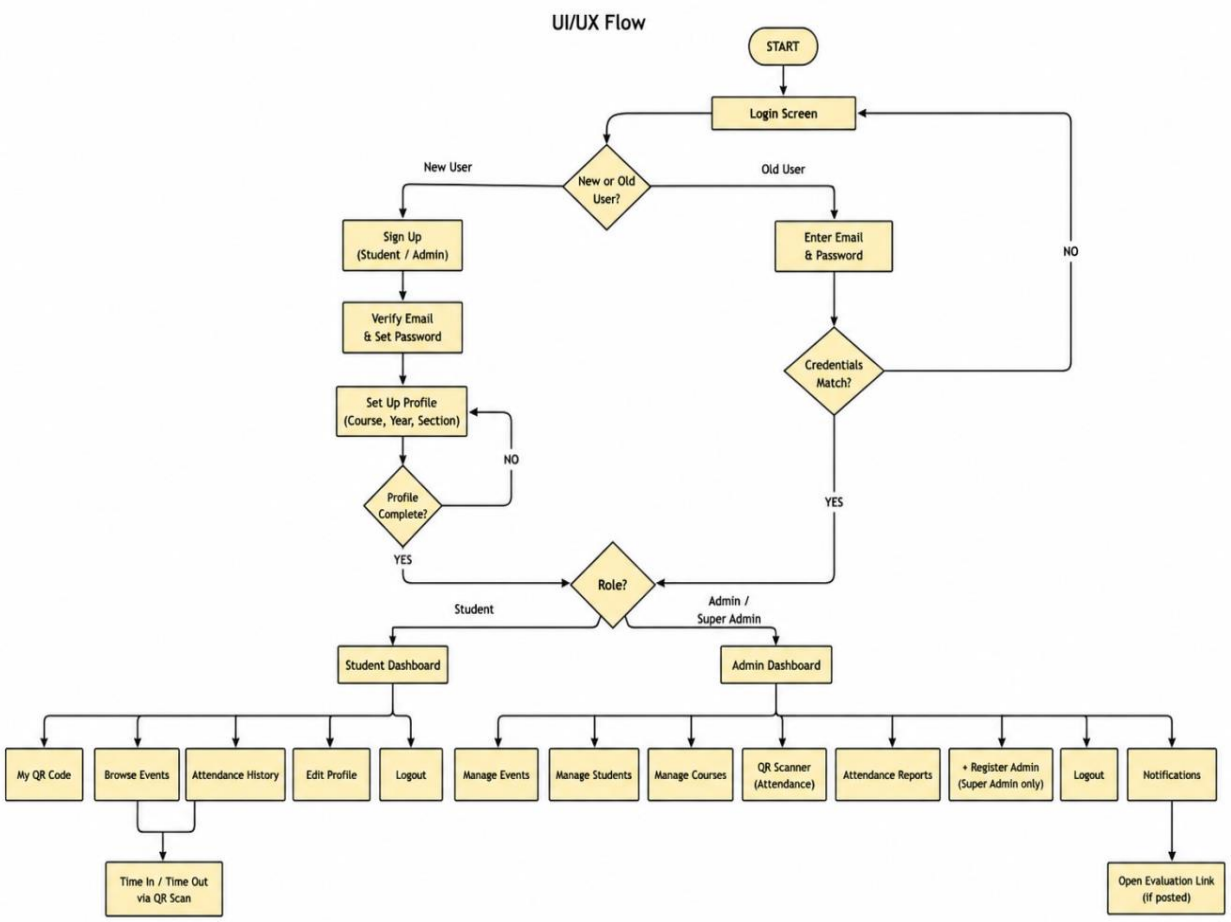


***Figure 12**: UI/UX Flow*

RELEASE NOTES

ATTENDTRACK: A Web-Based Event Attendance and Evaluation System in Lapu-Lapu City College

Version 1.0

Release Date: October 29, 2025

Introduction

AttendTrack Version 1.0 is the initial release of the web-based event attendance and evaluation system developed for Lapu-Lapu City College. This version was created to replace the traditional paper-based attendance process with a digital platform that records attendance through QR code technology.

The primary goal of this release is to provide administrators with an efficient way to organize events and monitor attendance while allowing students to participate through a secure and convenient attendance system.

Working Features

User Authentication

Secure login for administrators and students.

Role-based access to authorized system modules.

Password authentication and session management.

Event Management

Create and manage institutional events.

Set event schedules, venues, and descriptions.

Update and archive event information.

Student Management

Register student accounts.

Manage student profiles.

Store student information securely.

QR Code Attendance

Generate QR codes for attendance.

Record student attendance using QR code scanning.

Store attendance records automatically.

Attendance Records

View attendance history.

Search attendance records by student or event.

Maintain organized attendance information.

Dashboard

Display basic event information.

View attendance summaries.

Access system modules from a centralized dashboard.

Technical Details

Developed using Native PHP for backend processing.

Uses MySQL as the database management system.

Built with HTML5, CSS3, JavaScript, and Bootstrap for the user interface.

Runs on the Apache Web Server using the XAMPP development environment.

Designed as a browser-based web application.

Known Issues

Event evaluation is not yet available.

Event reminder functionality is not included.

Reporting features are limited to basic attendance records.

Dashboard analytics are limited in this version.

Limitations

Supports attendance recording only through the web application.

Requires an active internet connection.

QR codes are valid only during scheduled events.

Advanced reporting and analytics are not yet implemented.

ATTENDTRACK: A Web-Based Event Attendance and Evaluation System in Lapu-Lapu City College
Version 2.0

Release Date: December 15, 2025

Introduction

AttendTrack Version 2.0 introduces new functionalities that enhance the management of institutional events and attendance monitoring. This update expands the capabilities of the initial release by providing administrators and students with a more organized, efficient, and user-friendly platform.

The system now offers improved event management, QR code-based attendance, participant evaluation, event reminders, and centralized reporting. These enhancements reduce manual work, improve attendance accuracy, and simplify event administration.

Working Features

Event Management

Create, edit, archive, and manage institutional events.

Configure event schedules, venues, descriptions, and attendance settings.

Assign events to specific courses and departments.

Attendance Management

Generate unique QR codes for every event.

Record attendance through QR code scanning.

Display attendance records immediately after successful scanning.

Prevent duplicate attendance submissions.

Student Management

Register and manage student accounts.

Update student profile information.

View attendance history.

Display upcoming events assigned to students.

Event Evaluation

Allow participants to submit event evaluation forms.

Store evaluation responses.

Generate evaluation summaries for administrators.

Event Reminder System

Display reminders for upcoming events.

Notify students before scheduled activities.

Help students stay informed about event schedules.

Reports

Generate attendance reports.

View participant attendance summaries.

Produce event evaluation reports.

Dashboard

Display attendance statistics.

Monitor event participation.

View upcoming and completed events.

Access reports and system information.

Technical Details

Developed using Native PHP for backend processing.

Uses MySQL as the database management system.

Built with HTML5, CSS3, JavaScript, and Bootstrap for the user interface.

Runs on the Apache Web Server using the XAMPP environment.

Implements QR Code technology for attendance recording.

Known Issues

Event reminders are limited to in-system reminders only.

QR code attendance requires an active internet connection.

Large attendance reports may require additional loading time.

Limitations

Designed specifically for event attendance and evaluation at Lapu-Lapu City College.

Offline attendance recording is not supported.

Mobile application support is not available in this version.

QR codes are only valid during the scheduled event period.

ATTENDTRACK: A Web-Based Event Attendance and Evaluation System in Lapu-Lapu City College

Version 3.0

Final Release Date: January 28, 2026

Introduction

Version **3.0** represents the final and stable release of **AttendTrack**. This version completes all planned modules and delivers a fully functional web-based platform for managing institutional events, recording attendance using QR code technology, and collecting participant evaluations.

The system centralizes event scheduling, attendance recording, student participation, event reminders, reporting, and analytics into a single platform. It minimizes manual attendance processes, improves data accuracy, and enables administrators to monitor event participation in real time while keeping students informed through timely event reminders.

This release also introduces improvements in system performance, security, reporting, and usability, making the application ready for deployment within Lapu-Lapu City College.

Working Features (Version 3.0)

1. Administrator Features

Administrators have complete control over managing events, attendance, reports, and users.

Dashboard & Analytics

View event statistics and attendance summaries.

Monitor attendance trends and participation records.

Display recent system activities.

Generate attendance analytics by event, course, and department.

Event Management

Create, edit, archive, and delete events.

Manage event schedules, venues, and descriptions.

Generate unique QR codes for every event.

Assign events to selected courses and departments.

Student Management

Register and manage student accounts.

Update student information.

Activate or deactivate user accounts.

Manage student QR code records.

Attendance Monitoring

Monitor attendance in real time.

rch attendance records by event or student.

Prevent duplicate attendance entries.

View attendance history.

Evaluation Management

Create event evaluation forms.

Review submitted evaluations.

Generate evaluation summaries.

Monitor participant feedback.

Event Reminder Management

Schedule reminders for upcoming events.

Send event reminders to students before scheduled activities.

Manage reminder settings for each event.

Help students stay informed about event schedules and attendance deadlines.

Reports

Generate attendance reports.

Generate event evaluation reports.

Export attendance records.

View historical event data.

2. Student Features

Students can participate in events and monitor their attendance through a simple and user-friendly interface.

Student Dashboard

View upcoming and completed events.

Access attendance history.

View upcoming event reminders.

Monitor participation records.

Event Participation

Join assigned events.

Scan QR codes for attendance.

Verify attendance status.

View event information.

Event Evaluation

Submit event evaluations.

Provide comments and suggestions.

Review evaluation submission status.

Profile Management

Update profile information.

Change account password.

Manage personal account details.

3. General Features

Authentication & Access Control

Secure login and logout.

Role-based access control.

Session validation and account protection.

QR Code Attendance

Automatic QR code generation.

Secure QR code validation.

Duplicate scan prevention.

Instant attendance recording.

Event Reminder System

Upcoming event reminders.

Scheduled reminder alerts.

Attendance reminder notifications.

Reminder display on the student dashboard.

Reports & Analytics

Attendance summaries.

Event participation statistics.

Evaluation reports.

Dashboard analytics.

Completed Enhancements in Version 3.0

Complete QR Code Attendance Monitoring

Real-Time Attendance Tracking

Event Evaluation Module

Improved Dashboard Analytics

Attendance Search and Filtering

Attendance Report Generation

Enhanced QR Code Validation

Optimized Database Performance

Improved User Interface

Stronger Authentication and Security

Faster Attendance Processing

Improved Overall System Stability

Technical Details

Architecture: Three-Tier Architecture (Presentation Layer, Application Layer, and Data Layer)

Backend: Native PHP

Frontend: HTML5, CSS3, JavaScript, Bootstrap 5

Database: MySQL

Web Server: Apache HTTP Server

Development Environment: XAMPP

QR Code Technology: HTML5 QR Code Scanner

Authentication: Role-Based Access Control (Administrator, Super Administrator, and Student)

Deployment Platform: Browser-Based Web Application

Supported Browsers: Google Chrome, Microsoft Edge, Mozilla Firefox, and other modern browsers

Known Limitations

A stable internet connection is required for system access.

Offline attendance recording is currently unavailable.

The system is accessible through a web browser only.

QR code attendance is valid only during the scheduled event period.

Mobile application support is planned for future versions.

Version 3.0 Release Summary

Version **3.0.0** is the final stable release of **AttendTrack**. It includes all major system modules such as event management, QR code attendance, real-time attendance monitoring, event reminders, event evaluation, reporting, analytics, and role-based user management. This version provides a complete and reliable solution for improving event attendance monitoring and evaluation at Lapu-Lapu City College.

Deployment Plan

The AttendTrack: A Web-Based Event Attendance and Evaluation System (LLCC) has been successfully deployed as a web-based platform. Version 1.0.0 represents the final and stable release of the system, providing secure, efficient, and real-time management of event attendance, student records, and in-app notifications for the Lapu-Lapu City College community.

The deployment ensures reliable system performance, secure data handling, and seamless access for students, administrators, and super administrators through modern web technologies and industry-standard deployment practices.

Deployment Steps

1. Hosting

The system is hosted on a z.com Cloud Server environment that supports PHP 8.x and Apache HTTP Server, ensuring stable and efficient backend performance.

The hosting environment runs on a Linux distribution (Ubuntu 22.04 LTS) with the GD extension enabled for QR-image generation and the PDO MySQL driver for database access. The infrastructure is capable of handling concurrent users, simultaneous QR scans during events, and automated in-app notification processes without degradation of service.

2**. Domain and SSL Integration**

The system is deployed under a registered domain (e.g.,eventattendance.llcc.edu.ph) configured for public online accessibility. HTTPS with SSL certification has been enabled (via Let's Encrypt, auto-renewed by Certbot) to ensure secure data transmission across all pages. SSL encryption protects sensitive information such as student and admin credentials, session cookies, QR-code tokens, and attendance records. HTTPS is mandatory because the browser-based html5-qrcode scanner requires a secure origin to access the device camera.

3**. Database Migration**

The system utilizes MySQL 8.0 as its primary database management system. All database schemas, student records, admin accounts, event configurations, attendance logs, course catalogs, notifications, and analytics audit trails were deployed to the live environment from the local XAMPP build.

Data integrity and consistency are maintained through structured SQL migrations (database.sql, create_admin_tables.sql, create_event_departments_table.sql, student_notifications.sql, add_* migrations) and PDO prepared statements with transactional processing, ensuring accurate attendance and audit records.

4. **Feature Optimization**

Core system features were tested and optimized before final deployment, including:

Student and admin account management (signup, profile completion, QR code generation, account locking). Event creation and lifecycle management with department- and course-level invitation filters, evaluation links, and attendance lock controls.QR-based attendance capture (Time-In and Time-Out) via admin_qr_scanner.php using the html5-qrcode library. Administrative dashboards, attendance reports, and audit analytics powered by Chart.js 4.4.1 and exportable via jsPDF 2.5.1 + jspdf- autotable 3.5.31. In-app notifications (event reminders, time-in/out confirmations, evaluation link alerts) via the student_notifications table. Performance tuning (database indexes on QR codes, sessions, attendance, and notifications) was applied to ensure reliable system response and accurate real-time data updates.

5. Cross-Device Testing

The platform was rigorously tested for compatibility and responsiveness across various devices, including desktops, laptops, tablets, and smartphones. Modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge were used during testing to ensure consistent behavior and performance — including the camera-based QR scanner under HTTPS. The responsive design allows full system functionality (student QR display, attendance scanning, dashboards, and reports) without the need for a dedicated mobile application.

Future Plans

The system will undergo continuous monitoring and maintenance (uptime checks, log review, daily MySQL backups via mysqldump, and quarterly security patching) to ensure stability and security. Regular updates and enhancements will be implemented based on institutional needs and user feedback collected from students and administrators. Potential future improvements include SMS / e-mail notification gateways, multi-event timeout scheduling, role expansion (faculty advisers), advanced analytics with course-level attendance heatmaps, and scalability enhancements through containerization (Docker) for larger college-wide rollouts.

Configuration Guide

The AttendTrack system is designed using a modular, multi-component architecture to ensure secure operation, efficient attendance management, and real-time communication. The system consists of several key interacting components that work together to deliver seamless functionality.

1. Student Web Portal: This serves as the primary interface for students. It allows users to view their personalized QR code, browse upcoming events, complete their profile (course, year level, section), check their attendance history (Time-In/Time-Out), receive in-app notifications, and access the post-event evaluation link. The Student Web Portal communicates securely with the Backend Server through defined PHP endpoints to ensure accurate data retrieval and interaction.

2. Admin Panel: The Admin Panel is a web-based interface specifically designed for administrators and super administrators. It enables them to create and manage events (with department and course filters), manage student accounts and QR codes, maintain the courses catalog, scan student QR codes for attendance capture, lock or unlock attendance sheets, post evaluation links, and review attendance reports and analytics. Super administrators have an exclusive privilege to register new admin accounts via admin_signup.php. Changes made in the Admin Panel are immediately reflected across the system to maintain data consistency.

3. Backend Server: The Backend Server functions as the core processing unit of the system. Built on PHP 8.x running on Apache, it handles requests from both the Student Web Portal and the Admin Panel, manages business logic (QR generation, Time-In/Out rules, role-based access control via includes/admin_auth.php and includes/auth.php), enforces session security with bcrypt-hashed credentials and HTTP-only cookies, and dispatches in-app notifications. The backend ensures real-time synchronization of data and secure execution of system operations through PDO prepared statements.

4. Database: The database is responsible for securely storing and managing all system data, including student records, admin records, session tokens, event details, course/department invitation pivots, attendance logs (with Time-In and Time-Out), QR code metadata, in-app notifications, and analytics audit trails. The system uses MySQL 8.0 with the InnoDB engine and utf8mb4 character set to ensure reliability, data integrity (foreign keys with ON DELETE CASCADE / SET NULL), and efficient query performance through dedicated indexes.

5. Notification System: The Notification System interacts with the Backend Server to send automated in-app notifications to students. This includes attendance confirmations (Time-In and Time-Out), event reminders, evaluation-link availability alerts, and administrative announcements. Notifications are persisted in the student_notifications table and surfaced through the bell-icon badge on student pages. Delivery and read status are monitored (created_at, read_at) to ensure communication reliability and provide an auditable record of every alert issued by the system.

Technical Requirements

AttendTrack requires a web-based environment to operate efficiently and securely within an academic institution. The system is developed using standard web technologies and is designed to be accessible through modern web browsers without requiring additional software installation on client devices. It supports real-time attendance tracking, QR code scanning, and evaluation submission while ensuring centralized administrative control.

Hardware Requirements

To ensure stable and efficient system performance, the following hardware specifications are recommended:

Client-Side Hardware (User Devices)

Desktop computers, laptops, tablets, or smartphones.

Devices capable of running modern web browsers.

Camera-enabled devices for QR code scanning.

Stable internet connectivity for real-time access.

Android smartphones are primarily supported, but other platforms remain compatible.

Server-Side Hardware (Backend System)

Reliable server capable of continuous operation

Multi-core processor with at least 8GB RAM

Sufficient storage for system data, logs, and backups

Stable network connection for uninterrupted system access

Software Requirements

The system utilizes the following software components for development and operation:

Client-Side Software (User Interface)

The system interface is responsive and adapts to different screen sizes and devices.

Supported Web Browsers:

Google Chrome

Mozilla Firefox

Microsoft Edge

Server-Side Software (Backend System)

Programming Languages:

PHP 8.2

JavaScript

Database Management System:

MySQL (admin-managed database)

Web Server:

Apache (via XAMPP or similar local server environment)

Development Tools:

Visual Studio Code (or similar code editor)

QR Code Integration:

QR code generator and scanner libraries for attendance tracking

Operating System:

Compatible with Windows, Linux, or other standard server environments

Deployment Considerations

The deployment of AttendTrack focuses on implementing the system within an academic environment to support event management and attendance monitoring. At Lapu-Lapu City College. This phase ensures that the system is fully integrated, functional, and ready for institutional use.

Component Integration

All system components including the student interface, admin panel, backend server, database, QR code module, and notification system must be properly integrated to ensure smooth data flow and accurate attendance recording.

Real-Time Operations Optimization

The system is optimized to support real-time QR code scanning, immediate attendance validation (Time-In/Time-Out), and instant updates of attendance records and evaluation responses.

Data Management and Administration

The database is centrally managed by the system administrator, ensuring proper handling of student records, event details, attendance logs, and evaluation data. Regular backups and maintenance procedures must be implemented to maintain data integrity.

Security Implementation

Security measures include user authentication, role-based access control, encrypted passwords, and secure database transactions to protect sensitive information and prevent unauthorized access.

User Training and Adoption

Training sessions may be conducted for administrators and users to ensure proper system usage. These include guidance on event creation, QR code scanning, attendance monitoring, report generation, and system navigation.

USER GUIDE

AttendTrack User Guide

Introduction

Welcome to AttendTrack: A Web-Based Event Attendance and Evaluation System for Lapu-Lapu City College, a web-based platform designed to help administrators efficiently manage events, monitor attendance, and oversee student participation through a centralized dashboard with real-time data and QR code-based tracking.

This guide provides simple step-by-step instructions to help users navigate and use the system effectively.

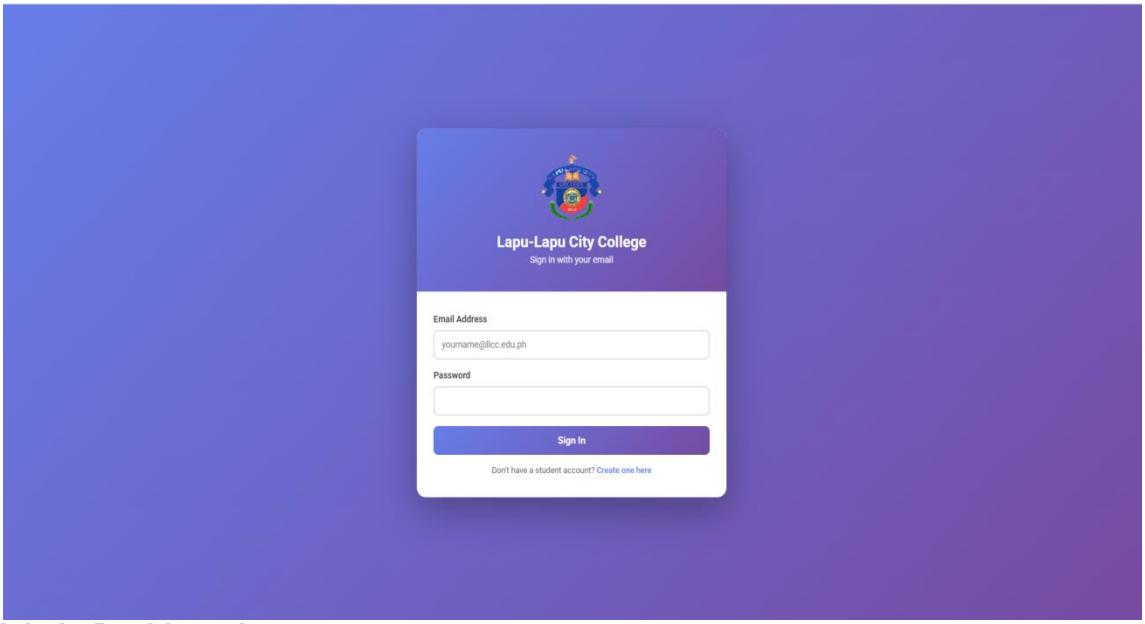
1. Admin User Guide

Role Overview: The Super Admin manages the whole system and controls global settings.

Getting Started

Log In

Enter your Admin Credentials



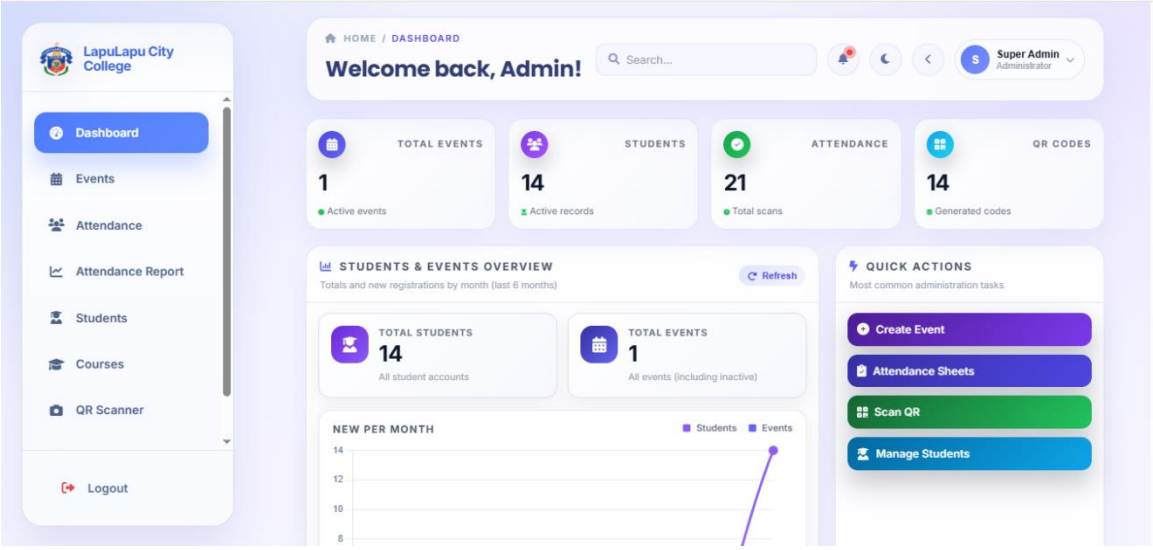
Admin Dashboard

Main overview of system statistics and activity

User Management

Add, edit, or remove users

Manage Role (Student, Event Organizer, Admin)



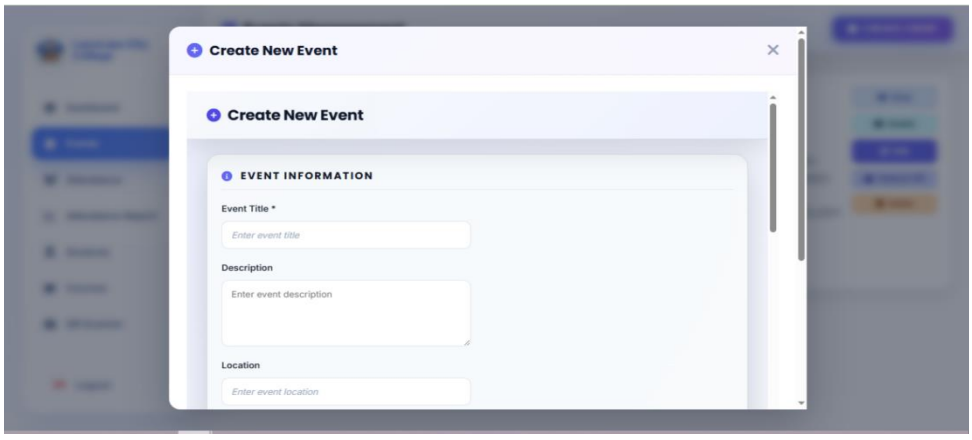
Pending Approvals

Review and approve

Verify and approve attendance logs

Reports

Request

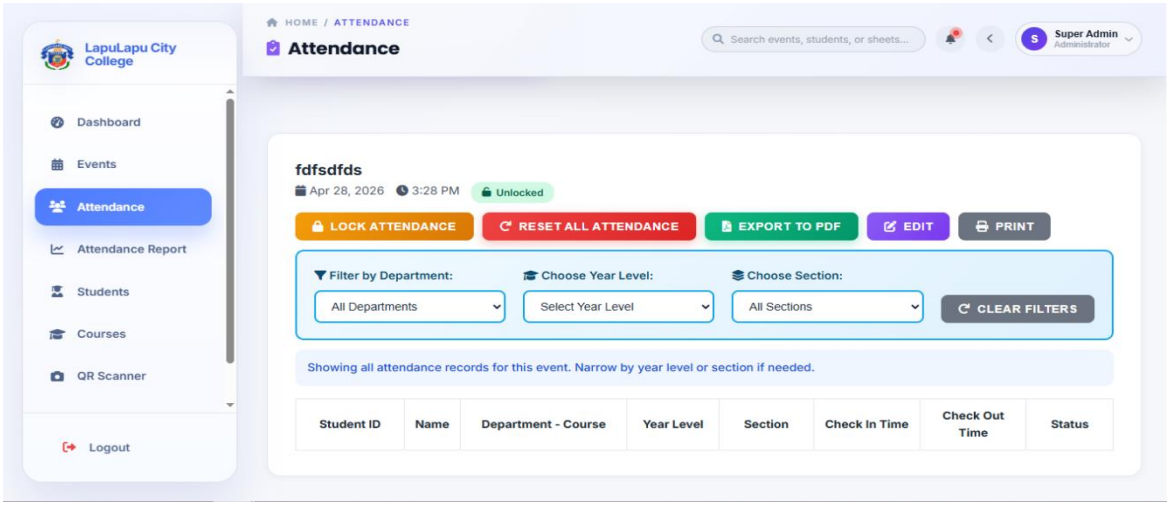


Event Management

Create new events (title, description, location, date, time)

Edit or delete existing events

Manage event details and participant information



Attendance Management

View and monitor attendance records for each event

Lock or unlock attendance to control submissions

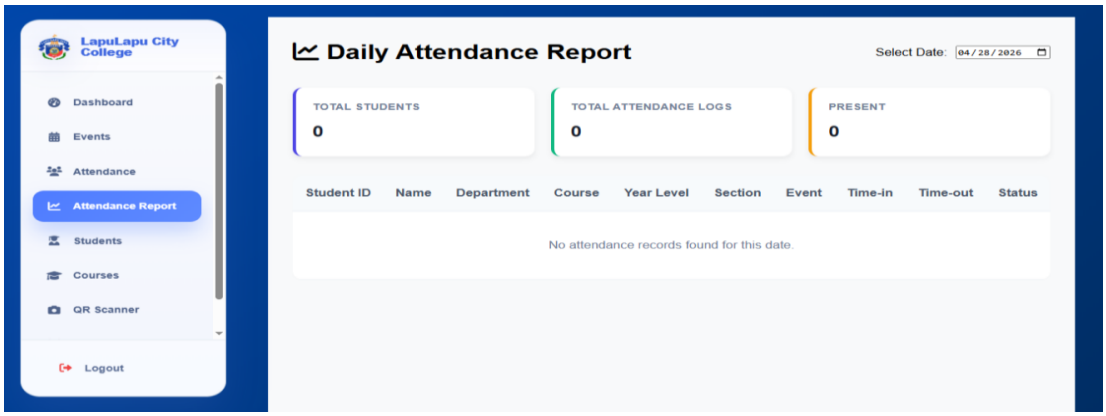
Reset all attendance records when needed

Export attendance data to PDF and print reports

Edit event attendance details

Filter records by department, year level, and section

Track student check-in and check-out times



Attendance Report

View daily attendance records based on selected date

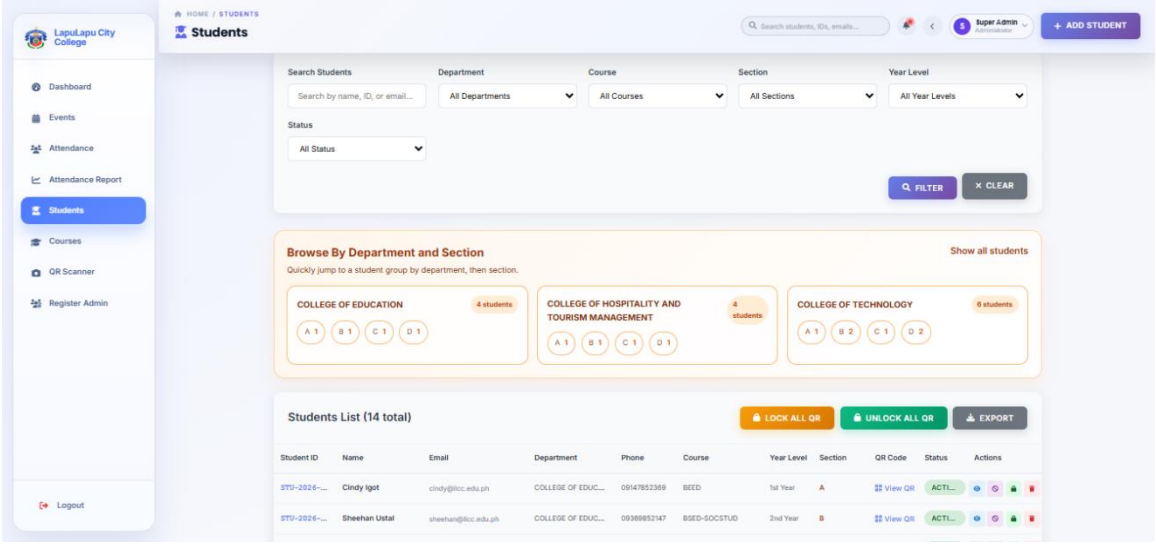
Display total students, attendance logs, and present count

Filter attendance using date selection

View detailed student information

Track attendance status

Generate and review attendance reports for monitoring and evaluation



Students Management

Search and filter students

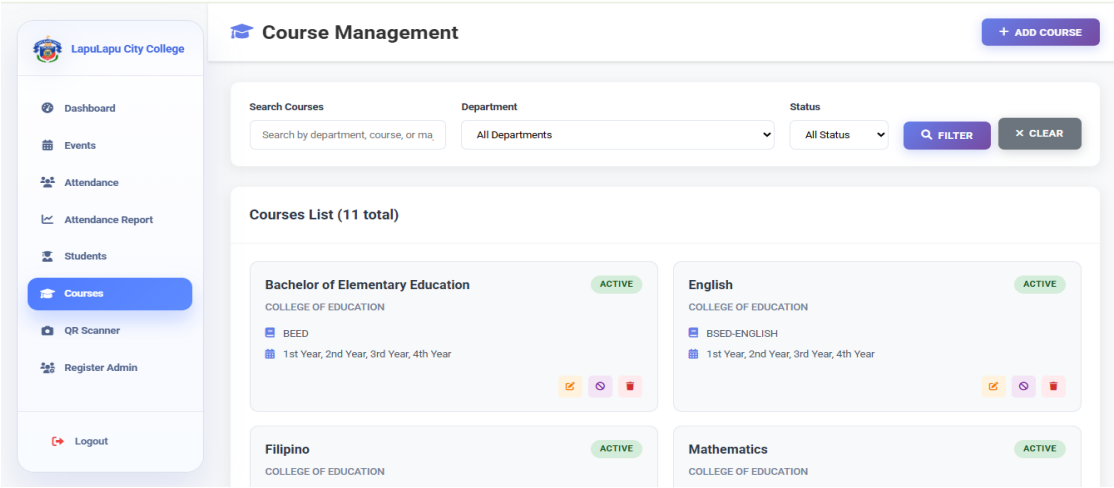
Browse by department and section

View student details and list

Add, edit, or delete students

Manage QR codes and status

Export student data



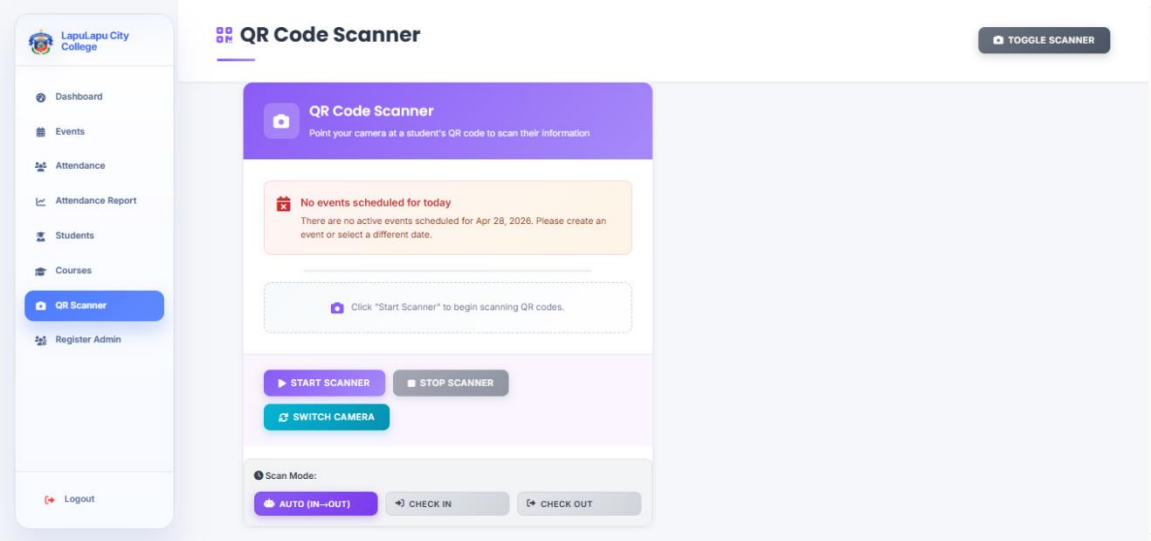
Course Management

Search and filter courses

View course list and details

Add, edit, or delete courses

Manage course status



QR Code Scanner

- Scan student QR codes for attendance
- Start or stop scanner
- Switch camera
- Select scan mode
- View scan status and event availability

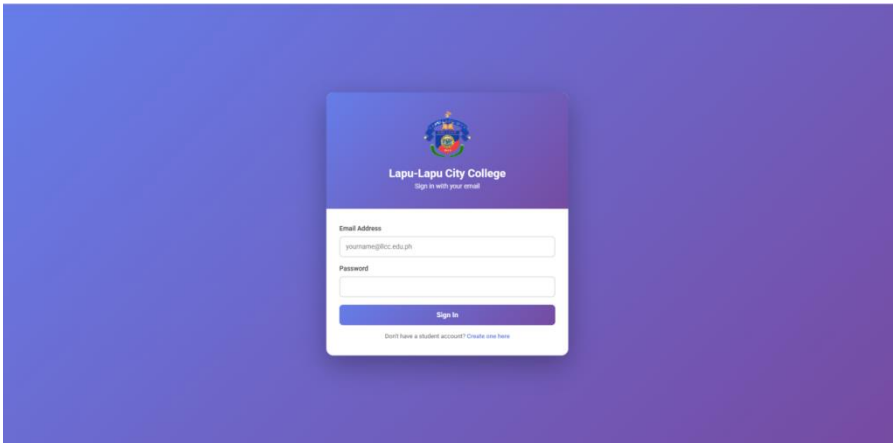
Student User Guide

The Student Dashboard provides a centralized and user-friendly interface that allows students to access event information, view personal details, and manage attendance within the AttendTrack system. It displays relevant upcoming events based on the student's course or department, along with essential profile information for verification. The dashboard also includes quick navigation to key features such as QR code access for attendance scanning and notifications for updates. Overall, it is designed to ensure efficient interaction, real-time access to information, and seamless participation in institutional events.

Role Overview:

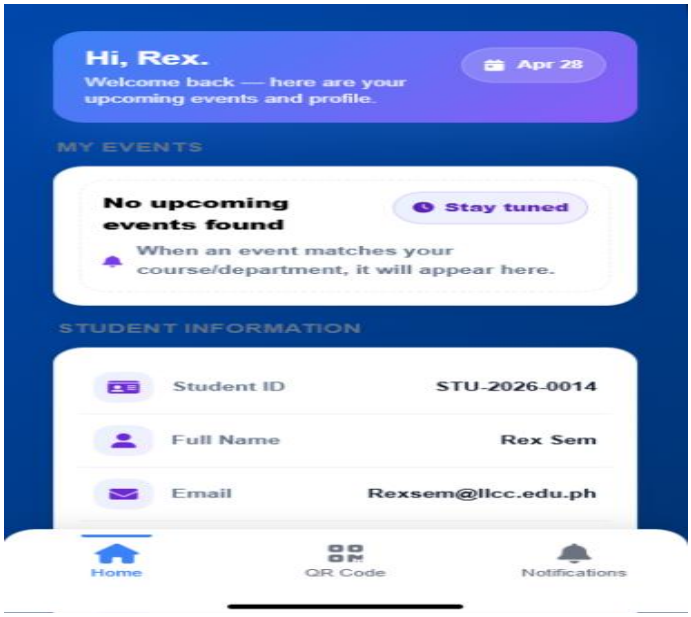
Log In

Enter your Student Credentials



1. Student Dashboard

Main overview of student activities



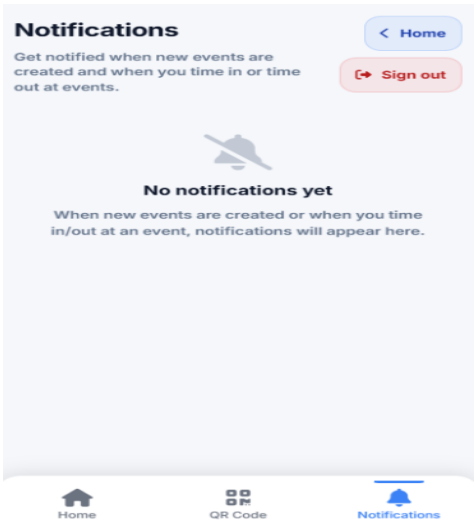
QR Code

View QR code Info and Download QR code or Printed



Notification

The system notifies students of new events and automatically triggers the evaluation form after time-out.



FUTURE RECOMMENDATIONS

Although the AttendTrack system has already brought significant improvements in inventory management operations, there are still areas that can be enhanced; here are some recommended system improvements for future development.

Develop a Mobile Application Version

Future developers may create a dedicated mobile application for Android and iOS devices to provide users with faster and more convenient access to event registration, attendance scanning, notifications, and evaluation forms without relying solely on a web browser.

Implement Offline Attendance Functionality

To address internet connectivity issues during events, future versions should support offline attendance recording. Attendance data may be temporarily stored on the device and automatically synchronized with the database once an internet connection becomes available.

Integrate Advanced Analytics and Reporting

Future enhancements should include interactive dashboards, graphical reports, attendance trends, participation statistics, and predictive analytics to assist administrators in evaluating event performance and making informed decisions.

Strengthen Security Features

Additional security mechanisms such as two-factor authentication (2FA), encrypted QR codes, automated audit logs, and advanced user authentication should be implemented to further protect user accounts and sensitive attendance records.

Support Multi-Institution Implementation

The system may be redesigned to support multiple schools, campuses, or organizations through configurable modules, allowing AttendTrack to be utilized beyond Lapu-Lapu City College with minimal customization.

Integrate Biometric Verification

Future researchers may combine QR code attendance with biometric authentication, such as facial recognition or fingerprint verification, to further minimize proxy attendance and improve attendance accuracy.

Expand Notification and Communication Features

Future versions may integrate SMS, email, and push notification services to automatically inform participants about upcoming events, attendance confirmations, schedule changes, and evaluation deadlines.

Conduct Large-Scale Performance Testing

Since the current study evaluated the system under a controlled environment with a limited number of respondents, future researchers should perform stress testing and load testing involving a larger number of simultaneous users to evaluate scalability, response time, and server performance.

Integrate with Existing Institutional Information Systems

Future development may include integration with student information systems, learning management systems, or institutional databases to automate participant registration, student verification, and event record synchronization while minimizing redundant data entry.

Conduct Longitudinal and Comparative Studies

Future researchers are encouraged to evaluate the long-term effectiveness of AttendTrack by conducting longitudinal studies and comparing its performance with other event attendance management systems implemented in different educational institutions. Such studies may provide broader evidence regarding the system's usability, efficiency, scalability, and overall impact on institutional event management.

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Appendix A

Transmittal Letter

April 20, 2026

Dr. Robert B. Pabillaran
Dean, College of Technology
Lapu-Lapu City College
B. Benedicto St., Gun-ob, Lapu-Lapu City, Cebu

Dear Dr. Pabillaran:

Greetings!

We respectfully seek permission to conduct a survey among the students, event organizers, and administrators of the College of Technology for our research entitled **"AttendTrack: A Web-Based Event Attendance and Evaluation System."**

The study aims to gather data on the current event attendance process, identify challenges in the existing manual system, and evaluate the proposed AttendTrack system. The collected information will be used solely for academic and research purposes.

All responses will be kept confidential and anonymous. Participation will be voluntary, and the information gathered will be used exclusively for this study.

We sincerely hope for your favorable approval. Thank you for your time and support.

Respectfully yours,

BSIT-3D Researchers

Degamo, Tristan A. **Quintero, Judel J.**

Perjes, Melchor S. **Ybañez, Reuven B.**

Noted by: **Approved by:**

MR. LINDO C. ESTRERA
Technology Research Adviser

DR. ROBERT B. PABILLARAN
Dean, College of Technology

Appendix B

Survey Questionnaire

Good day!

We are fourth-year Bachelor of Science in Information Technology (BSIT) students from Lapu-Lapu City College conducting a research study entitled "AttendTrack: A Web-Based Event Attendance and Evaluation System."

We kindly ask for your participation by answering the attached questionnaire honestly. Your responses will help us evaluate and improve the proposed system.

All information you provide will be kept strictly confidential and used only for academic and research purposes. By completing this questionnaire, you voluntarily agree to participate in this study.

Thank you for your time and valuable support. God bless

Directions: This questionnaire aims to gather information regarding the evaluation of the AttendTrack Web-Based Event Attendance and Evaluation System. Please read each item carefully and check (✓) the appropriate box that best represents your answer.

SECTION A: DEMOGRAPHIC PROFILE

1. Age

☐ 18–20 years old

☐ 21–23 years old

☐ 24–26 years old

☐ 27 years old and above

2. Role in the Institution

☐ Student

☐ Event Organizer

☐ Administrator

3. Level of Experience with Digital or Web-Based Attendance Systems

☐ No experience

☐ Limited experience

☐ Moderate experience

☐ Extensive experience

SECTION B: PROBLEMS IN THE EXISTING MANUAL EVENT ATTENDANCE SYSTEM

1. System Inefficiencies

☐ Manual attendance recording causes delays

☐ Slow verification of participant attendance

☐ Difficulty consolidating attendance data

☐ Increased administrative workload

2. Time Consumption Issues

☐ Time-consuming attendance tracking

☐ Slow preparation of attendance reports

☐ Delays in updating participant records

☐ Manual monitoring takes too long

3. Attendance Record Management Issues

☐ Incomplete attendance records

☐ Lost attendance sheets

☐ Encoding errors

☐ Duplicate attendance entries

4. Monitoring and Evaluation Issues

☐ Difficulty tracking participant involvement

☐ Lack of real-time attendance monitoring

☐ Difficulty evaluating event participation

☐ Limited access to attendance data

SECTION C: ROOT CAUSES AND SYSTEM GAPS

1. Technology and Automation Deficiencies

☐ Lack of automation in attendance tracking

☐ Manual data consolidation

☐ No QR code system

☐ No automated report generation

2. Centralized System Gaps

☐ Absence of centralized attendance database

☐ No unified event management system

☐ Data stored in multiple locations

☐ Limited access to historical data

3. Real-Time Monitoring Limitations

☐ No real-time attendance visibility

☐ Delayed updates during events

☐ Inability to track participation live

☐ No live reporting feature

4. Integration and Evaluation Gaps

☐ Attendance not linked to evaluation tools

☐ No integrated feedback collection

☐ Separate systems for attendance and evaluation

☐ Limited analytics for decision-making

SECTION D: EVALUATION OF THE ATTENDTRACK SYSTEM

1. Functional Suitability

Statement	4 (Very Functional)	3 (Functional)	2 (Less Functional)	1 (Not Functional)
The system automates attendance recording effectively.				
QR code scanning works accurately.				
The system generates complete attendance reports.				
Event evaluation features function properly.				
The system supports event monitoring requirements.				

2. Performance Efficiency

Statement	4 (Very Efficient)	3 (Efficient)	2 (Slightly Efficient)	1 (Inefficient)
Attendance recording is faster using the system.				
Reports are generated quickly.				
The system reduces manual workload.				
Real-time monitoring improves efficiency.				
System response time is satisfactory.				

3. Usability

Statement	4 (Very Usable)	3 (Usable)	2 (Slightly Usable)	1 (Unusable)
The system is easy to use.				
Navigation is clear and understandable.				
Attendance scanning is simple.				
Reports are easy to access.				
The system interface is user-friendly and visually clear.				

4. Reliability

Statement	4 (Very Reliable)	3 (Reliable)	2 (Slightly Reliable)	1 (Unreliable)
The system operates consistently.				
Attendance data is saved accurately.				
The system minimizes errors.				
Data is preserved after submission.				

5. Security

Statement	4 (Very Secure)	3 (Secure)	2 (Less Secure)	1 (Not Secure)
User login is secure.				
Access is limited to authorized users.				
Attendance data is protected.				
Unauthorized edits are prevented.				
Data privacy is maintained.				

Appendix C

GANTT CHART

No.	Feature	User Story	Task	Start Date	Start End
1	Account Management	As a user, I want to create an account and log in securely.	User registration, login, validation, and session control	Oct 1, 2025	Oct 6, 2025
2	User Dashboard (Administrator)	As an admin, I want a dashboard to view system summary and activities.	Dashboard layout, charts, and data integration	Oct 7, 2025	Oct 17, 2025
3	Event Management	As an admin, I want to create and manage events in the system.	Event CRUD, schedule setup, and details management	Oct 18, 2025	Oct 28, 2025
4	Participant Management	As an admin, I want to manage event participants and their information.	Add, edit, view, and remove participants	Oct 29, 2025	Nov 6, 2025
5	QR Code Attendance	As a staff, I want to record attendance using QR code scanning.	QR code generation, scanning, and attendance logging	Nov 7, 2025	Nov 17, 2025
6	Real-Time Monitoring	As an admin, I want to monitor live attendance during events.	Live attendance tracking and real-time updates	Nov 18, 2025	Nov 26, 2025
7	Notification System	As a user, I want to receive notifications about events and updates.	Email/push notifications and alert triggers	Nov 27, 2025	Dec 5, 2025
8	Evaluation Module	As a participant, I want to provide feedback and evaluate events.	Evaluation form, survey logic, and submission	Dec 6, 2025	Dec 14, 2025
9	Report Generation	As an admin, I want to generate reports for attendance and events.	Report filters, export options, and summaries	Dec 15, 2025	Dec 23, 2025
10	Analytics Dashboard	As an admin, I want to view analytics and insights.	Analytics charts, KPIs, and data visualization	Dec 24, 2025	Jan 2, 2026
11	User Role Management	As an admin, I want to manage roles and permissions.	Role setup, access control, and permission management	Jan 3, 2026	Jan 10, 2026
12	System Testing & Deployment	As a system owner, I want the system to be tested and deployed.	Testing, bug fixes, and system deployment	Jan 11, 2026	Jan 20, 2026

CURRICULUM VITAE

CURRICULUM VITAE



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Babag National High School	2022-2023
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Lapu-Lapu City College	2026-2027

SIMINARS/TRAINING/WORKSHOPS ATTENDED

Mental & Physical Health Awareness Seminar (2024 & 2025)

Gender and Development Seminar (2025)

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Mental & Physical Health Awareness Seminar (2024 & 2025)

Gender and Development Seminar (2025)

Certificate of Participation of Research (IT Days 2026)

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Gender and Development Seminar (2025)

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Religion: Roman Catholic

EDUCATION

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Bankal Elementary School	2013-2014
Secondary:	
Bankal National High School	2019-2020
Tertiary:	
Lapu-Lapu City College	2026-2027

SIMINARS/TRAINING/WORKSHOPS ATTENDED

Mental & Physical Health Awareness Seminar (2024 & 2025)

Gender and Development Seminar (2025)